Clicks and Bricks: The Complementarity of Online Retail and Urban Services

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Abstract

Is online retail a complement or substitute to local offline economies? To answer this question, I use transaction data and the staggered introduction of online grocery platforms. I find that online grocery delivery reallocates consumer purchases towards offline services through time savings, particularly restaurants. I also find that it changes how consumers make shopping trips, creating diverse spillovers. Consequently, online grocery delivery is both a complement and substitute to different offline stores. I develop a discrete choice model capturing nuanced trip choices which predicts the online grocery market will shift consumption from offline goods to services in high-income urban areas.

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1 Introduction

In the first quarter of 2024, U.S. household spending on online retail reached $289 billion, accounting for 16% of total retail sales (U.S. Census Bureau 2024). This significant shift towards online shopping is often portrayed as a "retail apocalypse" for physical stores. Indeed, brick-and-mortar retailers that are close substitutes to online retail naturally lose from its expansion (Dolfen et al. 2023). Widespread concern extends to related labor and real estate markets and to local governments' ability to generate sales tax revenue.[1] That said, online retail may complement other types of brick-and-mortar retail. For example, consumers can use their time saved from shopping online to frequent restaurants, salons, gyms, and other offline services consumed in-person. Existing research does not consider whether such complementary relationships between online and offline retail exist. Addressing this gap is crucial to anticipate the future of local economies and craft informed public policy responses to the growth of online retail.

In this paper, I study how online retail reshapes consumer shopping patterns to create both complements and substitutes and the resulting welfare implications for different consumer segments and brick-and-mortar stores. To do so, I first use credit and debit card transactions from 2012 - 2018 to construct detailed, high-frequency profiles of consumers' shopping behaviors, capturing online and offline transactions across locations and product categories. Second, I exploit the staggered rollout of online grocery delivery platforms, using transaction-based breakpoints to identify entry dates across ZIP codes and a difference-in-differences specification to identify their causal impact on spending and trips. Finally, I specify a discrete choice model of consumer trip choice, leveraging daily transactions, measures of neighborhood urbanization and demographics (including sex, age, income, and race), and individual-specific distance costs to estimate rich substitution patterns that complement the reduced-form results. This model quantifies the distribution of gains from online grocery delivery platforms and their impact on brick-and-mortar stores.

The reduced-form findings reveal multiple novel insights into the relationship between online retail and consumer shopping at brick-and-mortar stores. Contrary to the "retail apocalypse" narrative, I find that online grocery delivery increases spending and trips for offline services. The

[1] See, for example, Thompson (2014) on the decline of the retail workforce, Bhattarai (2019) on projected store closures, and Farmer (2019) on implications for local tax revenues.

increase I observe is concentrated in dine-in restaurants and coincides with an increase in spending on local ground transportation, suggesting that consumers reallocate time savings toward time-intensive services and travel in the local economy. Surprisingly, consumers who adopt online grocery platforms also *increase* their overall spending, consistent with online products facilitating substitution between time and money to create these offline complements. Additionally, consumer trip chaining, in which consumers visit multiple stores on a single trip to save on fixed and distance-related costs, can play a crucial role in creating complements and substitutes. For example, I find evidence that online grocery platforms reduce visits to both grocery stores and general goods stores on a chained trip because consumers no longer find it worthwhile to pay the fixed cost of shopping for general goods alone when they are also available online. In contrast, grocery store visits chained with offline services decline substantially less than grocery store visits alone or chained with goods. Therefore, trip chains that include visits to offline services offer a relative safeguard against competition from grocery delivery platforms.

Assessing the complex impact of online retail on consumer shopping behaviors involves tackling both data limitations and the challenge of causal identification. Comprehensive datasets that capture both online and offline purchases with sufficient detail to study heterogeneous spillover effects are scarce. For instance, the Consumer Expenditure Survey provides limited detail on online purchases, does not track specific stores visited on shopping trips, and includes too few households to detect the spillover effects of online retail purchases. Similarly, while smartphone data provide a good measure of physical trips at scale, they lack information on online purchases. The difficulty in establishing causality stems from unobserved factors that correlate with online purchasing decisions. For example, consider a consumer who increases their online clothing purchases while reducing spending at physical clothing stores and restaurants. It is challenging to determine whether the increase in online spending directly leads to the offline effects, or if an unrelated factor (like a new job or the birth of a child) drives spending patterns in both payment channels. These challenges have hampered research into the causal effects of online retail.

The transaction data I use are crucial for addressing these challenges in analyzing the emerging online grocery market. These data overlap with the period when many online grocery delivery platforms first entered new markets and are comprehensive enough to examine what was initially a niche market. During the 2010s, online grocery platforms entered many US cities in quick succession, commonly launching in high-income downtown ZIP codes before expanding into the suburbs. This pattern of expansion provides variation in access to online grocery delivery that is distinct from other determinants of individuals' consumption decisions. However, few customers initially used the platforms, so I focus on early adopters whose platform adoption was likely driven by the timing of platform entry. I compare their spending and trips before and after platform adoption against a matched comparison group of non-users from the same ZIP codes. The results are robust across alternative identification assumptions and are representative of effects for the broader population of online grocery platform adopters.

In the remainder of the paper, I examine the implications of the burgeoning online grocery market for consumer welfare and demand for different types of brick-and-mortar stores. To connect with the reduced-form findings, I employ the paired combinatorial logit (PCL) model, a discrete choice model that allows for differential similarity across pairs of choices (Koppelman and Wen 2000). In the model, I focus on consumers' choice of trips composed of binary combinations of a grocery store, restaurant, and general goods store to capture the effects of declining grocery store visits on goods and services. I allow preferences to vary flexibly with customer demographics and market urbanization to quantify heterogeneity in welfare gains and trip reallocation across consumers and locations. I then estimate the model using the same early platform adopters and matched non-users, allowing me to leverage the variation in platform entry to identify the model's parameters. Using these estimates, I find that consumer welfare gains in the early online grocery platform market average 2.2%.

I then estimate counterfactual outcomes for consumers and stores for a mature online grocery delivery market calibrated to the current adoption rate of 10%. Welfare gains accrue disproportionately to high-income consumers and those in the most urban ZIP codes. Young, female, and White consumers are also more likely to benefit. For brick-and-mortar stores, online grocery platforms on average decrease consumers' visits to grocery stores by 0.7% and increase their visits to restaurants and general goods stores each by 0.1%. However, these gains and losses are heavily concentrated in urban ZIP codes, with new restaurant visits more concentrated than those for general goods stores. Heterogeneity in changes in trip choices across consumers also suggests that online grocery platforms shift the customer base of in-person grocery stores toward lower-income and non-White consumers, while those for restaurants and general goods stores shift toward male, high-income, and White.

This paper's key contribution is causal evidence of the heterogeneous impacts of online retail on consumer consumption patterns, suggesting novel mechanisms that create both complements and substitutes in the offline economy. It builds on work exploring the impact of online retail on shopping behaviors across payment channels (e.g. Gentzkow (2007); Pozzi (2012)). Complementary research examines the substitution effect using online sales taxes and fulfillment centers (e.g. Goolsbee (2000); Ellison and Ellison (2009); Einav et al. (2014); Chava et al. (2023)). Other research shows that consumers gain from online retail through reduced search costs (Jin and Kato 2007), trade frictions (Jingting et al. 2018; Couture et al. 2021a), prices (Jo, Matsumura and Weinstein 2022), and increases in product variety (Quan and Williams 2018). Dolfen et al. (2023) emphasize variety and quality gains over reduced travel costs in welfare increases from the first two decades of growth in online retail.

By leveraging variation in access to online retail, this study also provides new reduced-form evidence of consumer substitution between time and money. To date, differences in consumers' opportunity cost of time generated by life-cycle and business-cycle changes have been used to study this substitution, crucial to foundational models of consumption and macroeconomic theory (Aguiar and Hurst 2007a; Aguiar, Hurst and Karabarbounis 2013; Nevo and Wong 2019; Bronnenberg, Klein and Xu 2023). My back-of-the-envelope calculation suggests that, for early adopters, access to online grocery platforms is equivalent to a 3.1% reduction in the price of online products.

This paper marks a significant development in modeling trip chaining by explicitly incorporating correlated utility shocks, which may arise when a single store appears in multiple multi-destination trip choices. I limit the model to the three major categories of stores to highlight the role of these correlated utility shocks in accurately depicting consumer substitution patterns. In doing so, this paper complements the work of Miyauchi, Nakajima and Redding (2025) and Oh and Seo (2023), who develop tractable models for the large choice sets for multi-destination trips. My results show that modeling correlated utility shocks is important to measuring the distribution of welfare gains and losses for consumers and retail firms. This methodology is widely applicable to a range of discrete choice problems where the independence of irrelevant alternatives assumption also falls short.

Finally, the results emphasize the growing consumption value of cities (Glaeser, Kolko and Saiz 2001). Recent research shows that the value of urban consumption density for consumers is large and based on non-tradable goods and services (Handbury and Weinstein 2015; Couture 2016; Cosman 2017; Davis et al. 2019; Couture et al. 2021b; Gorback 2022; Dingel and Tintelnot forthcoming; Almagro and Domínguez-Iino 2025). Consistent with this evidence, my findings suggest that online grocery delivery amplifies the consumption advantages of urban density. For retail firms, location decisions are shaped by spatial competition and the benefits of agglomeration (Houde 2012; Shoag and Veuger 2018; Qian, Zhang and Zhang 2024). I find that brick-and-mortar retailers in the most urban neighborhoods, particularly those specializing in services, will disproportionately benefit from positive spillover effects. Through these changes, cities can be a complement to online retail (Sinai and Waldfogel 2004).

2 Customer Transaction Data

I utilize proprietary data accessed through the JPMorgan Chase Institute containing credit and debit card transactions made by tens of millions of anonymized Chase customers starting in October 2012. Socioeconomic information available for each customer includes their sex, age, income, and race. Fields attached to each transaction determine payment amount, date, location, channel, and merchant. Locations for both customers and merchants are at the ZIP code level. I classify a transaction as online if the card was present at the time of purchase, the merchant is a known, large online-only retailer, or the store location is characterized by a website or phone number. Each transaction also carries a four-digit merchant category code which characterizes the primary good or service sold. I define a broader "product" categorization from these codes, enhanced by manually mapping a subset of leading merchants for online groceries and restaurants. Appendix section 6.A provides further detail about primary and secondary source data.

To study the spending behavior of individual customers, I construct a balanced panel of 7.2 million customers who make at least three transactions every month from October 2012 through April 2018. This subset of customers largely reflects the US demographic structure, though it skews somewhat male and high-income, reflecting a tendency for men to be primary account holders and low-income consumers to be unbanked. The sample is more urban than the population, reflecting Chase's geographic footprint.

In the customer panel, I focus on a subset of products that I define as "everyday products" – goods and services frequently purchased with cards in local markets. These include five goods-based categories: clothing, general goods, grocery, home maintenance goods and services, and pharmacy and four service-based categories: restaurants, coffee shops, local leisure goods and services, and personal care services.[2] Groceries, restaurants, and general goods account for nearly three-quarters of everyday spending offline. For online, general goods and leisure dominate, with the latter bolstered by recurring online payments for gym memberships and streaming services.

Patterns in the customer panel show broad scope for consumers to minimize the travel costs for offline trips via both single- and multi-destination trips. A large share of everyday purchases are at merchants in the same residential ZIP code of the customer. This is particularly true for groceries, for which 38% of offline transactions take place in customers' home ZIP codes. This pattern suggests that a consumer's home is a common end point for trips including these products and that consumers incur smaller travel costs for these products by purchasing them close to home. Other products, particularly services, are much more likely to be purchased outside the home ZIP code, implying more substantial travel costs. On days when consumers shop offline, more than half of those days include multiple offline purchases. On days with multiple purchases, consumers can chain many of those purchases together on the same trip, saving on travel costs by both shopping close to home and at stores that are close to each other. In this study, I consider all purchases made on the same day within a payment channel to be a single trip.[3] Descriptive figures summarizing spending and trips by payment channel are reported in appendix section 6.B.

[2]General goods include department stores, discount stores, large non-specific online retailers, and other miscellaneous retailers like florists and books stores that sell everyday goods. Major categories of personal care services include salons and dry cleaners. Major categories of local leisure include movie theaters and gyms. Other products are self-explanatory. Coffee transactions are identified through regular expression matching within the restaurant product category using major coffee chain names and words associated with coffee shops (e.g., "coffee" or "java").

[3]Unlike with smartphone data, I do not observe stays. Comparison with Miyauchi, Nakajima and Redding (2025) suggests that assuming all transactions that occur on the same day are on the same trip is roughly in line with defining trips by a sequence of stays that start and end at home.

3 Reduced-form Approach

The aim is to measure the causal effects of online retail purchases on consumers' other purchasing decisions. A major challenge in isolating these causal effects stems from the endogenous choices of the consumers and retailers. More specifically, consumers may experience shocks, like changes in employment or family composition, which change their preferred online and offline purchases. Concurrently, retailers might adjust their products and prices in response to the growth of online retail. These factors are often unobservable and difficult to control for, yet they can significantly influence both online and offline shopping behaviors. Therefore, simply observing the relationship between online shopping and other purchasing decisions is unlikely to reflect a causal effect.

I therefore utilize the unique features of the expansion of the online grocery market, which created variation in access to online groceries that was plausibly exogenous to individual consumption choices. In the following subsections, I discuss how this expansion is measured and characterized using transaction data and how it is applied in the empirical approach. I then discuss threats to identification, representativeness, and additional robustness checks. After presenting the main results, I explore time use and trip chaining as potential mechanisms.

3.A Expansion of Online Grocery Platforms

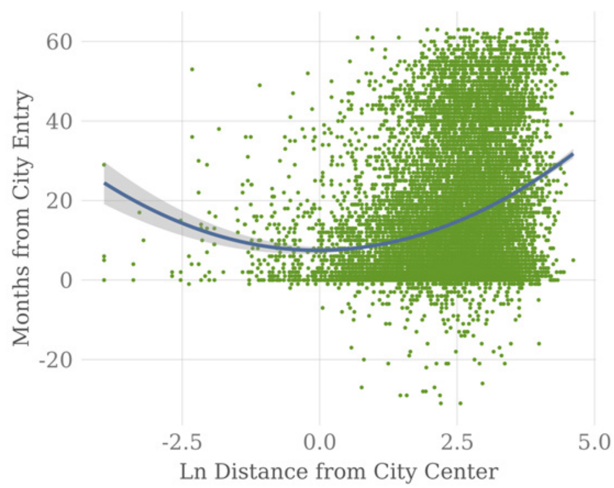
An online grocery delivery platform is a service that allows consumers to order groceries over the internet, which are then delivered to a specified location.[4] Platforms differentiate availability by the ZIP code of the delivery location. There is no public accounting of the dates at which different platforms were made available in different ZIP codes. Therefore, I utilize the transaction data of all customers to statistically pinpoint the month of entry of online grocery platforms into each city and ZIP code based on the breakpoint estimation methodology developed in Hansen (2000). While I cannot disclose specific platforms, I include the 17 platforms that I am aware of operating during my sample period. At entry, I observe a sudden and sharp jump in use that closely aligns with a

[4] I do not include services that offer a hybrid model of personal shoppers with customer in-person pickup. The recent expansion of such offerings would provide primarily in-store time savings and fewer fixed cost or travel time-savings. Grocery delivery remains the most popular form of online groceries (Saphores and Xu 2021).

subset of entry dates documented in news media. Appendix section 7 further documents platform entry measurement and customer platform use.

The decision of platforms to enter specific markets during the 2010s was influenced by several factors, most notably the pursuit of large urban customer bases. The many fixed costs of the online grocery delivery business, which could include refrigerated warehouses, a fleet of delivery trucks, and new drivers (e.g., Amazon Fresh), necessitated a significant, dense customer base. Some platforms entered more quickly through partnerships with or ownership of large existing brick-and-mortar grocery store chains (e.g., Instacart and Hy-Vee Aisles). Platforms also targeted cities near each other due to economies of density, as in Holmes (2011) and Houde, Newberry and Seim (2023).[5] In an interview, Instacart founder Apoorva Mehta also listed cities with poor weather, low car ownership, weak public transportation, and population age as key considerations in prioritizing city entry (Barr 2014).

Figure 1: Platform Expansion from Central to Suburban ZIP Codes



Notes: This figure shows the correlation between a ZIP code's distance to the city center and the timing of an online grocery platform's entry into that ZIP code. Distance is measured in miles and timing is measured as the number of months from the platform's entry into the city. The line is a quadratic fit of the relationship. See appendix sections 6.A and 7.A for the city center definition and details on estimation of ZIP code and city entry dates.
Source: Author's calculations using 98 billion transactions made by an unbalanced panel of 77 million customers from October 2012 through January 2020. Includes entry of 17 online grocery delivery platforms.

Within a city, as illustrated in Figure 1, platforms typically initiated service in a selection of central ZIP codes. Early entry was also more concentrated in ZIP codes with high population

[5]The entry pattern of Amazon Fresh is an example of this dynamic – after starting in Seattle in 2008, it expanded into Los Angeles and San Francisco in 2013; San Diego, New York, and Philadelphia in 2014; Baltimore and Sacramento in 2015; and Boston, Dallas, and Chicago in 2016 (Wikipedia 2020).

density and income. This approach allowed platforms to first target the most profitable areas, which had more potential customers and simpler delivery logistics. Then, they could expand upon these established routes to extend their reach into the suburbs. Platforms entering via an existing local grocery store chain also tended to provide more service immediately throughout the city. Roughly half of the ZIP codes in the cities where platforms launched eventually received service from a platform by January 2020.

Finally, the 2010s can be characterized as a scramble for early and rapid deployment of the online grocery platforms to as many attractive markets as possible. Online grocery platforms were a still new business model with many new entrants and earlier entry could bring advantages in market share. The focus on capturing early market share reduces the chance that entry timing was dictated by recent shifts in individual consumer spending.

3.B Estimation

I leverage the staggered expansion of online grocery platforms across ZIP codes within cities during the 2010s to identify the causal effects of online grocery delivery on consumer shopping behavior. A challenge of this approach is that the initial population-wide impact of entry is limited in most ZIP codes because few consumers used the platforms early on. At the end of the sample period, less than 1% of customers had transacted on one of the platforms. Moreover, there was low customer retention – 43% of those who tried a platform did not use that platform again in the next 18 months.

Therefore, I study the impact of online grocery platform entry on consumer consumption patterns using a difference-in-differences strategy focused on a select cohort: early adopters. I define early adopters as customers who start using their first online grocery platform within the first three months of its availability in their ZIP code (early) and continue using it in at least five of the first 18 months (adopter). In my sample, I observe a surge in platform adoption at entry. Among adopters, 22% are early, twice the share for temporary users. This concentration in adoption at the time of entry indicates that early adopters were eager to use the platforms at entry and would have adopted earlier if they were available. Therefore, unobserved factors that drive platform adoption are less likely to be correlated with the adoption *timing* for early adopters. Changes in consumption patterns at the time of platform adoption can be attributed to the effect of the platform entry, rather than confounding factors.

Early adopters of online grocery platforms differ from the general population. They are more likely to be female, younger, higher income, and spend more at grocery stores while going on fewer grocery trips. They also spend substantially more online on non-grocery products and travel more via local ground transportation prior to adoption. The challenge, then, is to identify a control group of non-adopting customers whose consumption of everyday products parallels the consumption of early adopters in the absence of platform entry.

For the control group, I select a matched group of customers with parallel pre-trends in consumption who never use a platform and live in the same ZIP codes as the adopters at the time of their adoption. This location requirement ensures that the control group faces the same unobserved neighborhood attributes, including local retail prices and store locations, but raises another concern. Why did these controls not use an online grocery platform if it was available to them? Early surveys showed that a majority of customers did not see the point of shopping for food online and were particularly reluctant to outsource selection of fresh produce (Mintel 2009). Differential exposure to early advertising and slow learning through social networks also determine early adoption of new technologies (Butters 1977; Conley and Udry 2010; Goolsbee and Klenow 2002). As long as these differences in preferences for in-person grocery shopping, information about platforms, or openness to new technology do not cause a violation of the parallel trends assumption, matched non-users from the same ZIP codes can serve as a valid control group.

To create these controls, I apply a standard propensity score matching technique (e.g. Dehejia and Wahba (2002); Caliendo and Kopeinig (2008)). I estimate a logistic regression where the probability of early adoption is modeled as a function of socioeconomic and consumption patterns prior to adoption:

$$\Pr(EA_{im}=1)=F\left(\mathbf{x}_i'\gamma+\mathbf{h}_{im-1}'\kappa\right), \tag{1}$$

where EA_{im} indicates that customer i is an early adopter in month m, $\mathbf{x}_i$ is a vector of time-invariant demographics capturing the customers' 2013 age, income, sex and race groups, and $\mathbf{h}_{im-1}$ is a vector capturing levels and changes for a subset of spending and trip history in the month prior to adoption. F is the logistic cumulative distribution function. Estimation of this equation delivers a propensity score for the likelihood that each customer is an early adopter of an online grocery platform. I then use the nearest-neighbor algorithm to pair each early adopter with two unique non-users without replacement from the same ZIP code whose propensity score is closest to the early adopter's.

I depart from the standard nearest-neighbor matching algorithm in several important ways. For computational ease, I do not match against all non-users in a ZIP code. Instead, for each adoption month and ZIP code combination, I randomly select a subset of non-users who live in the same ZIP codes at the time of adoption as potential controls. I then estimate the matching algorithm separately for each monthly group of early adopters, allowing the coefficients in equation 1 to vary over time. For instance, restaurant trip frequency might predict early adoption more in winter, while leisure trip frequency might do so in summer. My analysis includes early adoptions from April 2013 through January 2017, creating a balanced panel with two quarters before and four quarters after platform adoption. This approach matches 3,805 early adopters with controls.

My difference-in-differences strategy then compares early adopters against non-users from the same ZIP codes with parallel consumption trends prior to platform adoption at entry. Denote G_i as the quarter q of grocery platform adoption for customer i, where the first month of the quarter is the month of adoption. Then denote $D^l_{iq} = \mathbb{1}\{q - G_i = l\}$ as the relative quarter l of adoption dummy for customer i. I use the specification

$$Y_{im} = \alpha + \eta EA_i + \sum_{l=-2, l\neq -1}^{3} \lambda_l D^l_{iq} + \sum_{l=-2, l\neq -1}^{3} \delta_l D^l_{iq} EA_i + \mu_{im}, \tag{2}$$

where Y_{im} is the spending or trip outcome for customer i in month m and EA_i is an indicator for whether customer i is an early adopter. The δ_l are the coefficients of interest that measure the average effect of online grocery platform adoption on outcome Y_{im} for early adopters at quarter l. The key identification assumption is that factors causing early adopters to adopt at entry, while their controls did not, are uncorrelated with consumption trends for everyday products. Prior to adoption, δ_{-2} should be zero if the control group is a valid counterfactual. A positive δ_l in a post adoption quarter $0 \leq l \leq 3$ implies that platform adoption increases outcome Y_{im}.

I also measure the average change in the post-adoption year with the specification

$$Y_{im} = \alpha + \eta EA_i + \lambda D_{iq} + \delta D_{iq} EA_i + \mu_{im}. \tag{3}$$

where $D_{iq} = \mathbb{1}\{q - G_i \geq 0\}$.[6]

Further documentation of the matching design, covariate balance, and pre-adoption parallel trends is provided in appendix section 8.A.

3.C Threats to Identification, Representativeness, and Alternative Approaches

Despite this research design, it remains possible that unobserved factors correlated with both platform adoption at entry and everyday consumption patterns could bias the estimated coefficients. For instance, a subset of early adopters might have adopted an online grocery platform due to unobserved shocks that coincidentally align with the timing of a platform's entry into their ZIP code. Relatedly, small differences in trends for some pre-adoption consumption outcomes that were not used in the matching procedure remain statistically significant. Finally, using card transaction data means that changes in payment instruments could be correlated with the decision to use an online grocery platform.

A separate concern is whether the consumption response of early adopters is representative of the population. If these are the most enthusiastic adopters, effects on their consumption might be substantially larger than other consumers. Similarly, if they have distinct preferences or live in particular types of neighborhoods, effects may be qualitatively distinct.

To address these issues, I develop two alternative treatment and control groups. The first alternative broadens the treatment group to include "slower adopters" who adopt within the first year. In the data, I observe that initial use of online grocery platforms surges in the winter. This indicates

[6] Informed by Abadie et al. (2023), I show confidence intervals using classical standard errors. Concerns may arise about the partial clustering of treatment effects among individuals exposed to the same ZIP code and platform entry event. The panel data structure and matching individuals also implies potential benefits of clustering at the individual level. However, in this analysis, clusters (whether defined by ZIP code and platform entry or by individual) are relatively small and I find that clustering results in more conservative standard errors for the time-varying effects of adoption. Robust standard errors in this context are also found to be more conservative. The discrepancy between both clustered and robust standard errors is minimal and does not qualitatively alter the results. Finally, I do not include match-group fixed effects in the main specification, as identification is derived by construction from differences between treated and control units within each group (see appendix Table C4).

that customers are more likely to adopt during periods when weather conditions make in-person shopping unpleasant, which lowers the chances that their adoption timing is tied to demand shocks from family or employment changes. Extending the treatment group to those that adopt within a year of entry captures these customers whose adoption timing is determined by a combination of initial entry and seasonality. Controls for this treatment group are matched using the procedure described in section 3.B. The same identification assumptions apply, so comparing the treatment effects from this alternative to the main specification helps assess how representative the main effects are for a broader population.

The second alternative treatment group is the most expansive and includes all customers who adopt an online grocery platform, regardless of adoption timing. Rather than choosing from non-users, the potential controls for the "all adopters" treatment group are selected from temporary platform users (1-4 months of use in the 18 months following initial use) in the same ZIP code who use the same first platform in the same month. I match each adopter to a temporary user with similar demographics but a more limited set of shopping covariates because they are more alike before adoption. This approach shifts the identification assumption from exogenous adoption timing to post-adoption valuation of the platform's utility. The premise is that matched temporary users, initially as inclined to use the same platform at the same time as the adopter, find less value post-trial for reasons unrelated to pre-existing consumption trends. Therefore, identification remains valid as long as any unobserved shocks that correlate with adoption timing equally affect both adopters and temporary users.

Although the three treatment and control groups show level differences in average spending and trips for offline groceries, each comparison is made among groups with parallel pre-trends (appendix section 8.A). Notably, the all adopters treatment and control groups show increasing trends in offline grocery spending before platform adoption. This pattern implies that adopters whose timing is not driven by platform entry, either initially or in the first winter, are motivated by a substantial increase in grocery demand.[7] Unsurprisingly, there is minimal consumption of online groceries prior to platform adoption in each group.

[7] The divergent trends in pre-adoption use by early versus late adopters suggest that late adopters would violate the parallel trends assumption needed to serve as a control group for earlier adopters.

Post-adoption trends are also informative of whether each control group is a valid counterfactual for the treatment group. Controls for early and slow adopters maintain consistent trends in offline and online groceries after adoption. However, the all adopters control group shows a shift from flat to rising trends in offline grocery trips post-adoption. This suggests that these controls are able to increase their grocery consumption after discontinuing online platform use by making more frequent offline trips. In addition, while one-month triers revert to baseline usage of online groceries immediately after adoption, low users return more gradually. It is unclear whether these changes in grocery consumption would have occurred absent the trial use of the online grocery platform or if they are instead the result of shocks correlated with initial platform use.

Therefore, the best specification for causal identification depends on the extent to which each control group provides parallel trends in consumption before and after adoption. I prefer the comparison of early adopters and their matched controls as the main result because of the exogenous timing of platform entry and a control group that is unaffected by changes in platform availability. Nevertheless, the post-adoption counterfactual of temporary users could more accurately isolate the impact of online grocery platforms amid adoption-related shocks in the general population of platform adopters. Therefore, I present my main results using each alternative treatment and control group.

3.D Results

Figure 2 indicates that results from the three alternative treatment and control group estimations are remarkably consistent.[8] Thus, the main estimates reflect effects for a wider population of actual adopters. These effects matter because adopters shape how online groceries affect retail markets in aggregate. In the remaining, I focus on the results for early adopters, with effects denoted by circles in each panel.

In panel (a), I find that early adopters of online grocery platforms spend \$7.16 more per day on online groceries in the quarter of adoption. Over subsequent quarters, spending moderates as many early adopters do not use the platform consistently and some stop using the platform altogether.

[8] The main difference is that all adopters show a smaller relative increase in online restaurant spending and trips than the other treatment groups (Figure 2 panels (d) and (f)). This is not due to weaker post adoption spending by all adopters but to temporary users also increasing their online restaurant consumption after platform trial.

Still, over the subsequent year, online grocery spending remains elevated over $6.00 per day. On average, these consumers are making online grocery trips 2.2 more days per month in the year after adoption.

In response, early adopters of online grocery platforms significantly reduce their spending and trips at grocery stores after adoption. Panel (b) shows that prior to adoption, early adopters spend $22.85 per day at grocery stores and make grocery store trips 8.3 days per month.[9] In the quarter of adoption, consumers reduce their daily spending by $2.96 per day (-12.9%) and trips by 0.6 days per month (-7.5%). The decline in spending moderates somewhat over the following year, for an average decline of $2.65 per day (-11.6%). These reductions are the classic substitution effect for direct competitors to online retail. However, note that overall grocery spending after platform adoption increases by $3.85 per day (15.4%) in the year after adoption. This increase could reflect increases in overall grocery consumption, higher online prices, and delivery charges.

The spillover effects of this online grocery platform adoption go far beyond the impact on groceries. Panels (c) and (e) show that the adoption of online groceries leads to substantial increases in spending and trips for every major product of online retail except home goods and services. Across clothing, general goods, restaurants, and leisure, consumers spend an average $4.98 more per day online in the year after adoption. Thus, the adoption of online groceries constitutes a much larger shock to consumers overall online shopping.

Panels (d) and (f) show that there are disparate effects for offline goods versus services. Home goods and services show the most substantial decline of 0.4% in spending and 1.6% in trips in the year after adoption. Pharmacy is the only store for goods that is substantially positively impacted, increasing by 5.1% for spending and 2.2% for trips in the year after adoption. Offline services, by contrast, show across the board increases. Leisure has the strongest increase in spending in percent terms, 9.1%, followed by personal care, 7.4%. But, the 4.1% ($0.73 per day) average increase in spending at offline restaurants in the year after adoption is the most economically meaningful. Total spending across these eight offline products increases by $1.49 per day after adoption.

[9] These features of in-store and online grocery shopping are consistent with industry statistics on grocery shopping trends (Food Marketing Institute 2019). The primary grocery shopper for a household makes around 1.5 grocery store trips per week (6.5 trips per month). Among online grocery shoppers who purchase groceries online every month, online groceries constitute only one-third of their grocery expenditures.

To appreciate the broader impact of online grocery platform adoption on overall consumption, Figure 3 shows the total effect of platform adoption on online and offline spending and trips for early adopters. These results include products other than the nine everyday products. Panel (a) shows that total online spending increases by an average of \$15.40 per day (12.0%) in the year after adoption. In contrast, panel (b) shows that total offline spending declines by \$0.81 per day (-0.8%). The net effect of platform adoption is to increase total spending by \$14.58 per day (6.4%).[10]

3.E Mechanisms

In this section, I present evidence of mechanisms that create complements and substitutes among non-grocery products. I focus on two main mechanisms. The first is changing time use, which is significant because surveys show that time savings drive predominantly high-income and female shoppers to use online grocery platforms (e.g., Morganosky and Cude (2000)). These savings can increase consumption in service categories that require in-person time. The second mechanism is trip chaining, which can reduce demand for unrelated goods by altering the relative travel costs of shopping trips. Other mechanisms, such as income effects and goods complementarity, may also play a role but are left for future research. Additional figures referenced in this section are in appendix section 8.B.

In examining both mechanisms, I study variation in trip distances as a way to understand changes in travel behavior following online grocery platform adoption. I calculate distances using the ZIP code centroids for the customer's residence and the offline card terminal(s). Chained distances for visits to multiple stores are computed by first arranging stores chronologically by hour and then geographically by ZIP code, and then determining centroid-to-centroid distances. For trip segments within the same ZIP code, I approximate the travel distance as half the diameter of a circle with an area equivalent to that of the ZIP code.

Time use: For context, consumers spend around 5.2 hours per week purchasing all goods and services (including in-store and online purchasing), around an hour of which is travel-related (U.S. Bureau of Labor Statistics 2019). How much time might be saved by online grocery delivery in my sample? To understand the magnitude of the potential travel savings from grocery store trips, I

[10] I find no meaningful change in use of cash or checks at platform adoption for early adopters (Figure C5).

treat every grocery store visit as an individual round trip to the consumer's home and measure the change in the cumulative distance of those trips each month. Figure C6 shows a modest 2.7 miles per month, or 2.8%, decline in the distance traveled by consumers for grocery store visits. This decline is small because consumers primarily shop for groceries close to home. If consumers walk to the store at a rate of 3 mph, they would save 54 minutes each month.

Other time costs of grocery shopping can also be substantial. An analysis of the American Time Use Survey by the USDA suggests that, around the start of my sample period, the national average for time spent on grocery in-store shopping per day for those that shopped was 44 minutes (Hamrick et al. 2011). Coupled with the finding of 0.6 fewer grocery trips per month, this suggests a reallocation of 26 minutes from in-store shopping time every month. In sum, adopters of online grocery platforms likely recuperate roughly 1.3 hours monthly from grocery shopping, not counting the potential time saved from reduced trips to purchase other goods or fixed costs of in-person grocery shopping.[11]

I find evidence for time use as a mechanism in other concurrent changes in spending and trips at adoption. Figure C7 shows that spending and trips for local transportation increase by 11.9% and 11.2%, respectively. This suggests that not only are more trips overall being made, but that the new trips do not require the transport of heavy goods (unlike groceries). I also show in Figure C8 that the extra spending at restaurants is almost entirely through spending in eating and drinking places, as distinct from catered establishments (i.e. cafeterias) and fast food. The fact that spending increases substantially more than trips at eating and drinking places could indicate additional spending per trip that comes with spending more time (e.g., staying for dessert) or substitution toward time-intensive, higher-end establishments.

Trip chaining: To understand changing patterns in trip chaining, I study the frequency with which trips for each everyday product are made without a grocery store versus chained with a grocery store. The results presented in Figure 4 indicate that the decline in trip chains for goods purchased with grocery is the primary explanation for the decline in visits for clothing, general goods, and home goods and services. Each of of these chained trips declines substantially, while trips alone

[11]Fixed costs of in-person grocery shopping can be substantial: list preparation (online platforms typically offer reordering from previous shopping lists), managing children, and handling groceries. Some of this time is necessarily reallocated toward online grocery shopping. Anesbury et al. (2016) find that mean time for a 12-item basket is 11.3 minutes for inexperienced online shoppers.

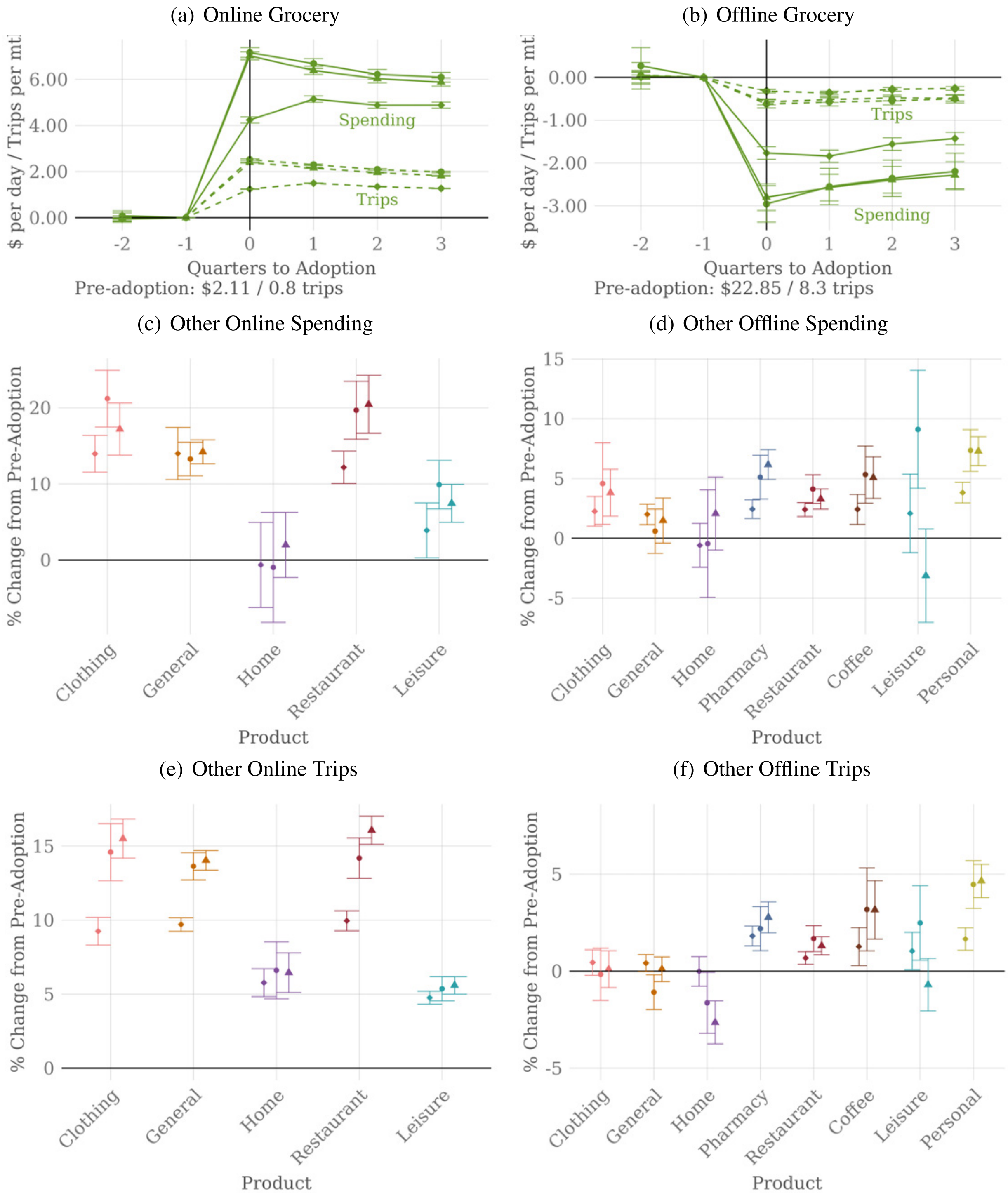
for these products increase. Thus, consumers appear to no longer be willing to pay the fixed cost of an offline shopping trip for goods alone when goods are easily available online. In contrast, services, which do not have the same online availability, have generally smaller declines in the trips that are chained with grocery. Indeed, that fact that they are not easily accessible online may insulate grocery stores from larger declines in trips.[12]

I also find that platform adoption affects the cumulative miles traveled by consumers across all trips (Figure C11). I perform this calculation under both the assumption that each transaction represents a single round trip from home and the assumption that transactions on the same day are part of a chained trip. I find that monthly average single trip mileage increases by 17.8 miles, or 2.6%, after platform adoption. However, chained trips show miles traveled increases by only 7.4 miles, or 1.5%. This discrepancy suggests that integrating trip chaining could diminish the distance-related costs of the newly adopted travel patterns by almost two-thirds. Additionally, the relatively lesser increase in distance measured by chains as opposed to single round trips suggests that consumers are forming new trip chains over shorter distances than before, concentrating trips closer to home locations. Assuming all new trips are made using local transportation at a speed of 12.5 mph implies 0.6 - 1.4 hours of additional travel time each month for trips that are fully chained versus single-store round trips.[13] This time range is of the same order of magnitude as the estimated 1.3 hours saved from in-store grocery shopping, which is consistent with the view that adopters reallocate saved grocery time to other local trips.

[12]Figure C9 is consistent with time reallocated from distant grocery store visits to restaurants to non-grocery stores near consumers' homes. Figure C10 shows that weekday versus weekend effects are most pronounced for chained trips. Consumers particularly decrease their chained trips for grocery with goods products more dramatically on weekdays, when time budgets for large trips are likely tightest.

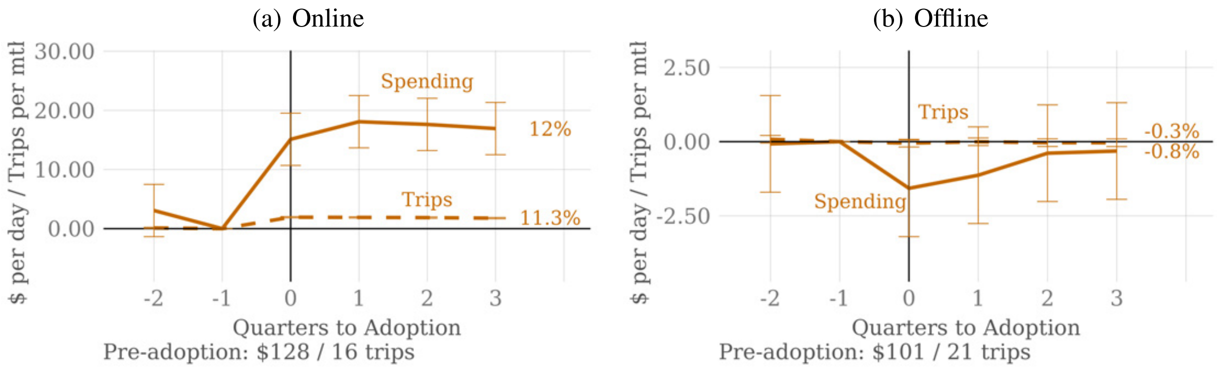
[13]Calculation uses the 2013 average speed of local buses reported in American Public Transportation Association (2015).

Figure 2: Everyday Spending and Trip Patterns at Adoption



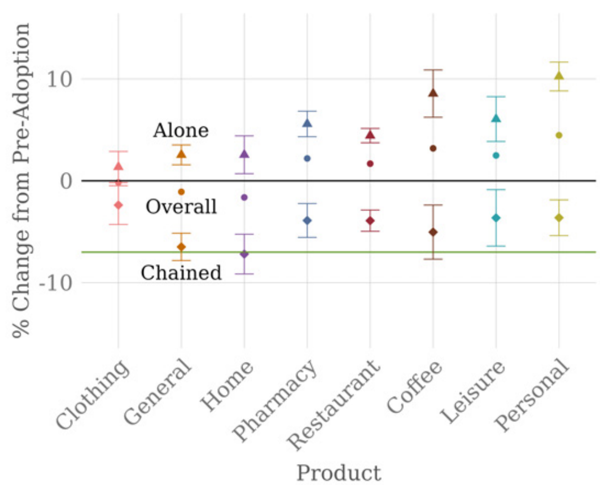
Notes: The top panels show the change in spending (solid lines) and trips (dashed lines) for online and offline groceries in the quarters before and after platform adoption using three alternative treatment and control groups. Circles denote early adopters, triangles for slow adopters, and diamonds for all adopters. Average outcomes for early adopters in the quarter prior to adoption are displayed below. Panels (c)-(f) show the average post-adoption effects as a percentage of the average value in the quarter before adoption. Figures C3 and C4 show the event studies.
Source: Author's calculations using the transactions of 11,418 early adopters of online grocery platforms and their matched controls from a set of platform non-users, 22,541 early and slow adopters and their matched controls from a set of platform non-users, and 55,831 adopters (regardless of adoption timing relative to entry) and their matched controls among platform triers and low-users.

Figure 3: Total Spending and Trips at Adoption



Notes: This figure shows the total change in online and offline spending (solid lines) and trips (dashed lines) in the quarters before and after online grocery platform adoption for early adopters. Averages for the quarter prior to adoption are displayed below each subfigure. The average percent effect for the year after adoption relative to the quarter prior to adoption is displayed at the end of the series.
Source: Author's calculations using the transactions of 11,418 early adopters of online grocery platforms and their matched controls from a set of platform non-users.

Figure 4: Changes in Trip Chaining with Grocery Stores



Notes: This figure shows the percent change in trips for products when chained with and without grocery after online grocery platform adoption for early adopters. Trips are defined as all offline transactions that take place on the same day. Coefficients are for the percent change for each product trip chained with grocery (diamonds) and without grocery (triangles) relative to the overall effect for that product (circles). The horizontal green line is a reference for the overall decline in grocery trips after adoption. Figure C9 shows additional changes in store combinations on trips.
Source: Author's calculations using the transactions of 11,418 early adopters of online grocery platforms and their matched controls from a set of platform non-users.

4 Model of Trip Choice and Platform Adoption

In this section, I develop a model that accounts for the interaction between substitutes and complements in consumer shopping trips and platform adoption. I model a nested choice in which a consumer first makes a long-run decision to adopt an online grocery platform or not and then, conditional on that choice, which shopping trip to make each day. An important contribution is advancing a model of trip choice which replicates the rich substitution patterns evident in the previous empirical exercise. To do so, I focus on trip combinations of grocery stores, restaurants, and general goods stores, as these categories account for most spending on everyday products. This focus captures how shifts in grocery shopping behavior differentially affect goods and service sectors. The model is solved by backward induction, with the consumer first determining the expected value of her daily trip choices conditional on her long-run platform adoption decision.

Throughout, I assume a fixed supply side and abstract from general equilibrium effects. Use of online grocery platforms was low during this period, so platform entry was unlikely to induce meaningful short-run changes in location choices, store entry and exit, or prices. The estimates are best interpreted as partial equilibrium effects conditional on prevailing market conditions.

4.A Daily Trip Choice

To start, I consider consumers indexed by i who are ex ante identical apart from their location. The market has one grocery store, one restaurant, and one general goods store. The consumer's utility maximization problem for the stores to include on a trip each day is

$$\max_{g,r,n} V_i(g,r,n) = \beta_0(g,r,n) + \tau \ln(d_i(g,r,n)) + \varepsilon_i(g,r,n), \tag{4}$$

where $V_i(g,r,n)$ is the trip value based on the store set $\{g,r,n\}$, with g,r, and n taking binary values for the grocery store, restaurant, and general goods store, respectively. The consumer has eight possible shopping trips with, for example, $V(g,r,n) = V(1,1,0)$ representing the value of the trip to the grocery store and restaurant. Store values for the trip are

$$\beta_0(g,r,n) = G\mathbb{1}_{\{g=1\}} + R\mathbb{1}_{\{r=1\}} + N\mathbb{1}_{\{n=1\}} + b(g,r,n), \tag{5}$$

where G is the grocery store value, R is the restaurant value, N is the general goods store value, and $b(g,r,n)$ is a fixed benefit for the trip chain of a particular multi-store trip combination. The fixed benefit for visiting more than one store can reflect non-distance benefits of trip chaining (e.g., leaving home once) and other complementarities in joint consumption as in Gentzkow (2007). Trip utility is moderated by τ, the opportunity cost of time, times the log distance, $d_i(g,r,n)$, traveled on each trip in miles.[14] The consumer has a random taste shock, $\varepsilon_i(g,r,n)$, for each trip.

The central theoretical challenge in this setting is modeling the correlation across individual consumer trip choices within a day. Trip values for a single consumer are often correlated due to factors like time constraints or the complementarities between stores. For example, a consumer who prefers a restaurant on a given day is more likely to select trips that include the restaurant, creating positive correlations across trips that incorporate it.

Adding to this complexity are the distinct trip distance costs faced by each consumer, as highlighted in studies of retail markets for gasoline (Houde 2012), restaurants (Couture 2016), and grocery stores (Thomassen et al. 2017). In this context, where consumers may visit multiple stores on a single shopping trip, the entire menu of distance costs shapes trip choices, not just the distance to an individual store. For example, a consumer living closer to a restaurant may be more likely to visit the restaurant alone, while another living near a grocery store and general goods store might chain a trip to both locations due to lower marginal travel costs for chaining.

To address these challenges, I use the paired combinatorial logit (PCL) model, a tractable approach from the transportation literature used to model trips (Prashker and Bekhor 1999; Koppelman and Wen 2000; Wang et al. 2017). Within the generalized extreme value class from McFadden (1978), the PCL model allows me to incorporate individual-level data with distinct distance costs and flexibly captures cross-elasticities between pairs of choices. The PCL model is well-suited for this context because it accommodates both substitutes and complements, does not require preconceived notions of choice order or nests, and allows each pair of choices to have a similarity relationship that is distinct from other pairs. Further highlighting the model's ability to capture complex substitution patterns, Li and Webster (2017), Zhang, Rusmevichientong and Topaloglu

[14] I discuss the choice of functional form for distance costs in appendix section 9.C

(2020), and Compiani, Morozov and Seiler (2023) are examples which apply the PCL model to product selection.

With a change of notation for concision, denote the eight possible trip choices $\{g,r,n\}$ as $grn \in \{000,001,010,011,100,101,110,111\}$. In the model, the probability of trip grn depends on the probability of trip grn relative to trip grn', $P_{grn|grn,grn'}$, and the probability of a pair of trips, $P_{grn,grn'}$ across the possible $grn \neq grn'$. The total probability of a trip grn sums over the product of these two terms for all trips grn',

$$P_{grn} = \sum_{grn' \neq grn} P_{grn|grn,grn'} P_{grn,grn'}, \tag{6}$$

where

$$P_{grn|grn,grn'} = \frac{\exp\left(\frac{V_{grn}}{1-\sigma_{grn,grn'}}\right)}{\exp\left(\frac{V_{grn}}{1-\sigma_{grn,grn'}}\right) + \exp\left(\frac{V_{grn'}}{1-\sigma_{grn,grn'}}\right)},$$

$$P_{grn,grn'} = \frac{\exp((1-\sigma_{grn,grn'})I_{grn,grn'})}{\sum_{grn}\sum_{grn' \neq grn}\exp((1-\sigma_{grn,grn'})I_{grn,grn'})},$$

and

$$I_{grn,grn'} = \ln\left[\exp\left(\frac{V_{grn}}{1-\sigma_{grn,grn'}}\right) + \exp\left(\frac{V_{grn'}}{1-\sigma_{grn,grn'}}\right)\right].$$

$I_{grn,grn'}$ is the inclusive value of trip pair $\{grn, grn'\}$ and $0 < \sigma_{grn,grn'} < 1$ is a similarity index for two trips, with $\sigma_{grn,grn'}$ increasing in similarity. Because there are $\binom{8}{2} = 28$ possible pairs of trips, there are 28 similarity parameters, denoted $\boldsymbol{\sigma}$. This model has a similar structure to the traditional logit and nested logit models, but with more flexible substitution patterns. It does this by using a logit shock over the value of a trip within a trip pair combined with a logit shock for the value of the trip pair. This allows simultaneous nesting of any trip with every other trip to account for different degrees of substitution between each pair of trips. As a result, trip grn is more likely if trip grn is valuable relative to trip grn' or the value of the pair of trips $\{grn, grn'\}$ is high (due to high trip utility parameters in the two trips or high similarity between them). The PCL model is equivalent to the logit model when the similarity parameters are all zero. In appendix section 9.B, I provide detailed formulations and illustrations of the substitution patterns under different model assumptions using the parameter values I estimate.

4.B Trip Choice Parameter Identification

I utilize the same data and identification strategy from section 3 for early adopters and their matched controls to estimate trip utility parameters before and after platform adoption and the trip pair similarity parameters. For tractability, I represent the trip choice set by assigning each consumer a set of "preferred stores" for each trip based on their typical shopping patterns before adoption. For each of the eight trip types, I take the distance cost of each trip as the median distance of realized trips for each customer prior to online grocery platform adoption (or counterfactual adoption in the case of non-users).[15] I assume a distance equivalence of traveling to no store on the none trip of 0.1 miles, reflecting the need to incur some cost to consume at home (e.g., cooking). In implementation, trip utilities also depend on a vector $\mathbf{x}_{zi}$ that captures variation in preferences for each trip based on the population density of the consumer's residential ZIP code z and socioeconomic characteristics of age, sex, income, and race. This leads to an expanded trip utility function,

$$
\begin{aligned}
V_{it}(g,r,c) = &\beta_0(g,r,c) + \beta_1(g,r,c)Post_t + \beta_2(g,r,c)EA_i + \beta_3(g,r,c)Post_t \times EA_i \\
&+ \tau \ln(d_i(g,r,c)) + \mathbf{x}_{zi}'\boldsymbol{\nu} + \varepsilon_{it}(g,r,c).
\end{aligned}
\tag{7}
$$

Narrowing the trip choice problem to preferred stores changes both the identifying variation and the interpretation of the coefficients. The vector $\boldsymbol{\beta}$ are trip-specific utility coefficients identified by trip choice frequencies. The intercept terms $\beta_0(g,r,c)$ identify average store values of G, R, and N through the probabilities of trips that visit each store alone when travel costs are zero. The fixed benefits of chained trips, $\mathbf{b}$, are identified from the differences in probabilities of chained trips

[15] For example, if the median round-trip straight-line distance for a customer for a restaurant alone is 5 miles in the twelve months prior to adoption, I take their trip cost for any restaurant alone to be 5 miles. Similarly, median round-trip straight line distances for chained trips are calculated using the implied chained distance traveled on days when multiple stores are visited. I restrict the sample to those consumers who choose each of the eight trip types at least twice in the year prior to adoption and whose median distance traveled on those trip types is less than 100 miles. I need at least one observation of each trip type to measure the separate distance cost of each type of trip. I require two to reduce the effect of extreme outliers. Large median distances occur when consumers live far from their reported ZIP code or spend substantial time away.

relative to single store trips. As in equation 3, $Post_t$ controls for post adoption differences in trip frequency that are common to all customers, while EA_i controls for constant differences between early adopters and their controls over the sample period. The interaction $Post_t \times EA_i$, with coefficients $\beta_3(g,r,c)$, captures changes in trip frequency after platform adoption and identifies changes in average store values and trip chaining benefits when consumers shop online for groceries. The distance parameter τ now measures the impact of trip distance across trip types at preferred stores rather than the opportunity cost of time in a model with full store choice. Finally, substitution across trips at different preferred trip distances provides identifying variation for $\boldsymbol{\sigma}$. Changes in trip preferences after adoption, holding other factors fixed, generate additional exogenous variation for $\boldsymbol{\sigma}$.

I estimate the log of equation 6 with trip utility defined by equation 7 via full-information maximum likelihood.[16]

4.C Platform Adoption Choice

I separately model the choice to be an online grocery platform adopter as a simpler, long-term choice. The choice depends on the value the consumer receives from the online grocery platform and the expected value of the daily trips that she will make conditional on that choice. I set the consumer's utility maximization problem for online grocery platform adoption as

$$\max_{O_i} U_i = (G_z^p + \mathbf{x}_i'\boldsymbol{\theta})\mathbb{1}_{\{O_i=1\}} + I_T(O_i) + \mu_i(O_i), \tag{8}$$

where U_i is the long-term utility of the consumer, G_z^p is a set of ZIP code fixed effects that reflect the value of the online grocery platform in ZIP code z, $\mathbf{x}_i$ is a vector of consumer socioeconomics influencing grocery platform value, $I_T(O_i)$ is the inclusive value of the set of offline trip pairs as a function of online platform use, and $\mu_i(O_i) \sim$ EV type 1 is the taste shock for being an online grocery shopper. Underlying platform values in each ZIP code can vary based on the quality and quantity of available services. Intuitively, platform adopters are those consumers who have

[16] In settings with many zero market shares, the literature in industrial organization and urban economics recommends estimation by maximum likelihood, the use of moment inequalities for shares, or models that explicitly account for zero market shares (Quan and Williams 2018; Dubé, Hortaçsu and Joo 2021; Gandhi, Lu and Shi 2022; Dingel and Tintelnot forthcoming). Aggregate trip frequencies for the eight trip types are shown in Figure D1.

a higher value for an online grocery platform and the trip pairs they would choose as an online grocery shopper as compared to the value of the trip pairs they would choose otherwise. Formally, consumers who adopt an online grocery platform have

$$G_z^p + \mathbf{x}_i'\boldsymbol{\theta} + \mu_i(1) - \mu_i(0) > I_T(O_i = 0) - I_T(O_i = 1)$$
$$= \ln\Big[\frac{\sum_{grn}\sum_{grn' \neq grn} \exp((1-\sigma_{grn,grn'})I_{grn,grn'})\,(O_i = 0)}{\sum_{grn}\sum_{grn' \neq grn} \exp((1-\sigma_{grn,grn'})I_{grn,grn'})\,(O_i = 1)}\Big]. \tag{9}$$

The total welfare gain from the platform, as measured by compensating variation, equates to the log sum gain in value from the platform.

$$\Delta CV_{zi} = \ln\Big[\frac{\exp(G_z^p + \mathbf{x}_i'\boldsymbol{\theta} + I_T(O_i = 1)) + \exp(I_T(O_i = 0))}{\exp(I_T(O_i = 0))}\Big], \tag{10}$$

which reduces to a simple function of the probability of online grocery platform adoption,

$$\Delta CV_{zi} = \ln\Big[\frac{1}{1 - P_{O_{zi}}}\Big], \tag{11}$$

where $P_{O_{zi}}$ is the probability of adoption of the online grocery platform for consumer i in ZIP code z. Intuitively, the higher the probability of adoption, the higher the increase in welfare to the consumer from being able to be an online grocery shopper. Thus, platform values rationalize the leftover variation in adoption rates observed in the data in each ZIP code after consumer socioeconomics and offline trip values.

I estimate equation 8 via maximum likelihood using all customers in the constant panel for whom I can calculate median offline trip distance costs for each trip type using their trips in 2013. Geographically, I also restrict this analysis to ZIP codes with at least 500 of these customers and at least one who adopts a platform during the sample period to focus on ZIP codes where there are sufficient customers to identify grocery platform value in the ZIP code.

4.D Estimates

Trip utility and similarities: The parameters mapped from the estimation are in Table 1 and Figure 5. Starting in Table 1 panel (a), the base values for each trip are values for the omitted demographic category (male, lowest income group, Black) when continuous variables are normalized to zero. Note that the pre-adoption store values are ranked $N < G < R$. This follows from the trip frequency ranking of each trip to an individual store relative to the none trip, with a value normalized to zero. In addition, the fixed benefits of chained trips, the $\boldsymbol{b}$, are positive for all but one of the chained trips, reflecting that visiting multiple stores on the same trip lowers the combined trip cost relative to visiting each store on separate trips. These benefits can be substantial – increasing chained trip values by as much as a third. Post-adoption, the estimates show a decline in the grocery store's value and the fixed benefits of chained trips that include the grocery store. The values for the restaurant and the general goods store slightly rise, with a higher increase in the fixed benefit of their chained trip.

The model also estimates the effect of distance on trip choice given customers' preferred stores for each trip. The τ parameter in panel (b) is an order of magnitude smaller than the effect of distance on store choice in choosing among all stores estimated from other trip choice models. This makes sense if consumers' choice of which kind of trip to make each day to their preferred stores is primarily determined by their overall demand for those products rather than distance. The more notable difference is that τ is positive, indicating a preference for distant trips within customers preferred stores for each trip. This is likely driven by consumers who frequently visit restaurants being more likely to visit farther restaurants for gains from variety (Couture 2016; Su 2022a).[17]

Panel (c) shows that more urban locations and consumer socioeconomics substantially shift consumer trip preferences. In urban areas, trips including the general goods store are less preferred while trips to the restaurant and grocery store alone and in combination are more preferred. Age most strongly increases the frequency of trips to general goods stores and decreases trips to restaurants alone. Across all types, female consumers make fewer trips and higher-income consumers make more trips. Finally, there is considerable heterogeneity by race in trip preferences; generally, for instance, Black consumers make fewer of most trips and Hispanic consumers do more trip chaining.

[17] A previous version of this paper without restaurants estimated $-0.01 \leq \tau \leq -0.02$.

Table 1: Trip Utility Parameters

(a) Store Values

	Pre-adoption	*Post-adoption*
G	-2.04	-2.15
R	0.06	0.08
N	-4.20	-4.18
$b(110)$	0.55	0.49
$b(101)$	-0.34	-0.52
$b(011)$	1.21	1.30
$b(111)$	2.26	2.20

(b) Distance Coefficient

τ	0.04

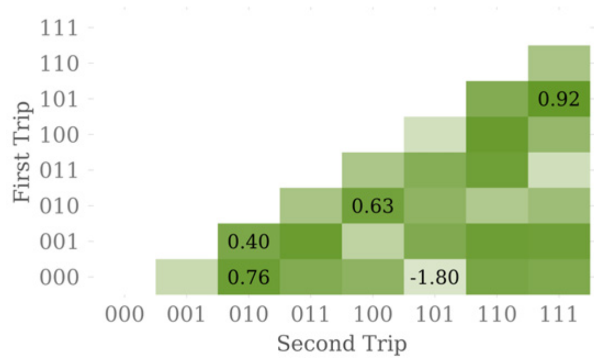
(c) Density and Socioeconomic Shifters

Trip grn =	*001*	*010*	*011*	*100*	*101*	*110*	*111*
ln Density	-0.05	0.01	-0.06	0.04	-0.06	0.01	-0.07
ln Age	0.36	-0.24	0.09	0.08	0.53	-0.16	0.13
Sex F	-0.03	-0.38	-0.37	-0.22	-0.31	-0.46	-0.48
Sex U	0.04	-0.21	-0.12	-0.01	0.06	-0.17	-0.16
Income 2	0.10	0.20	0.29	0.21	0.54	0.51	0.75
Income 3	0.29	0.30	0.49	0.33	0.98	0.69	0.99
Income U	0.15	0.12	0.19	0.26	0.65	0.47	0.66
Asian	-0.11	0.19	0.06	0.27	0.46	0.44	0.41
Hispanic	-0.06	0.34	0.36	0.11	0.15	0.50	0.70
White	-0.01	0.20	0.17	0.35	0.61	0.50	0.58
Race U	0.06	0.14	0.17	0.24	0.67	0.39	0.53

Notes: Trip utility parameters mapped from model estimates with trip utilities as specified in equation 7. Table D2 shows the estimates with standard errors. Pre-adoption values of stores derive from the trip value for that store alone, e.g. $N = \beta_0(001)$. Fixed benefits of trip chains pre-adoption derive from differences in values of standalone store visits and chained trips, e.g. $b(110) = \beta_0(110) - G - R$. Post adoption values incorporate β_3 coefficients.
Source: Author's calculations using the transactions of 6,764 early adopters of online grocery platforms and their matched controls.

The estimated similarity parameters are shown in Figure 5. They range in value from 0.92 for the trip pair for the grocery store chained with the general goods store and all stores on a single chain, implying nearly perfect similarity, down to -1.80 for the trip pair for the none trip with

Figure 5: Similarities

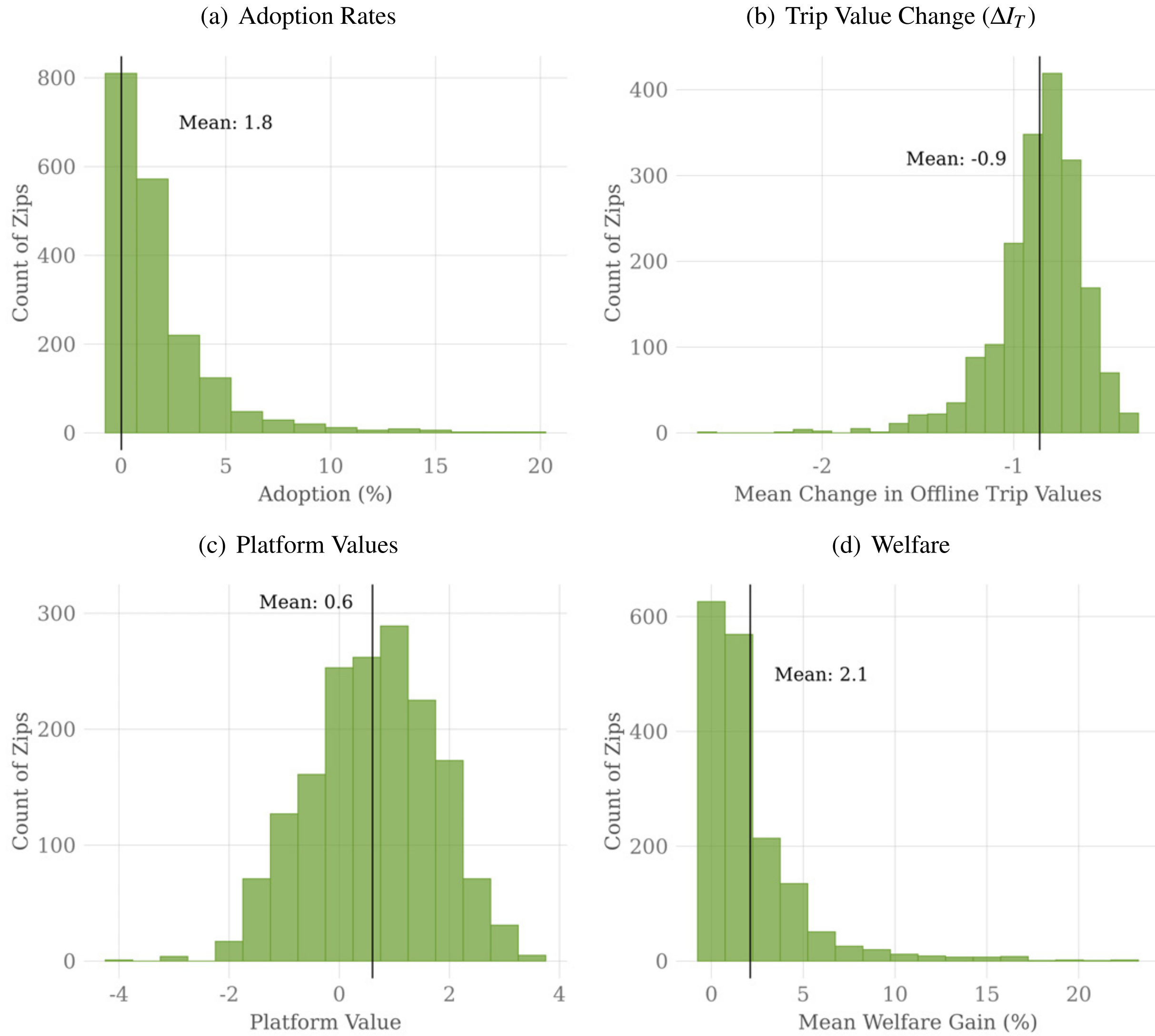


Notes: This figure shows the trip pair similarities for the PCL model of trip choice. Darker shading indicates higher trip pair similarity. The eight possible trip choices $\{g,r,n\}$ as $grn \, \varepsilon \, \{000,001,010,011,100,101,110,111\}$ where g, r, and n are the binary choice of grocery store, restaurant, and general goods store, respectively. See Table D2 for all point estimates and standard errors.
Source: Estimates derived using the transactions of 6,764 early adopters of online grocery platforms and their matched controls.

the chained trip to the grocery and general goods store, implying low similarity for the trip pair. Three additional key trip pairs are highlighted because they are similar and frequent. For example, the similarity between grocery alone and restaurant alone is 0.63, indicating strong substitution towards restaurant trips when the grocery alone trip value decreases. The high similarity between restaurant alone and no trip is also high (0.76). Together these capture the intuition from the reduced from: consumers prefer substituting goods stores for services when goods stores are less valuable, but will choose to stay home instead if the values of the service trips are too low.

Grocery platform adoption: Figure 6 panel (a) shows wide cross-ZIP code variation in adoption, with a mean of 1.8%. Panel (b) reports the ZIP code-level average change in trip value with adoption; the mean of these ZIP code-level averages is -0.9%. This cross-sectional variation, combined with consumer demographics, identifies the ZIP code fixed effects G_z^p and taste shifters θ in equation 8. Figure 6 panel (c) and Table 2 present the estimated distribution of G_z^p and the θ coefficients. Estimated platform values are small and often negative, consistent with low adoption in most ZIP codes. The largest socioeconomic associations are with age and income. While the average welfare gain across all consumers computed using equation 11 is 2.2%, panel (d) shows there is similarly wide distribution of the ZIP code-level average welfare gain from adoption. The mean of these ZIP code-level averages is 2.1%.

Figure 6: Grocery Platform Adoption



Notes: This figure shows derivations toward estimating online grocery platform value by ZIP code. Panel (a) shows the distribution of adoption rates. Panel(b) shows the distribution of within-ZIP code means of the model-derived decline in value of offline trips when consumers adopt an online grocery platform. Panel (c) shows the distribution in values for the platforms implied by the adoption rates and offline trip values using equation 8. Panel (d) shows the distribution of within-ZIP code average welfare gains from online grocery platform adoption estimated via equation 11.
Source: Author's calculations using 1,690 ZIP codes with at least 500 customers from the constant panel of customers and at least one customer who lives there and adopts a platform.

I note that a bit more than half of consumer gains from platform adoption reflect ZIP code-specific platform values. Furthermore, platform adoption reduces consumers' value for offline trips, which dampens realized gains. If trip values were held fixed at pre-adoption levels, average gains would be about four times larger. This comparison highlights that, by construction, the

Table 2: Predictors for Individual Platform Adoption

Dependent Variable: Platform Adoption				
Ln Age	Sex F	Sex U	Income 2	Income 3
-1.6	-0.15	-0.24	1.1	1.8
(0.04)	(0.01)	(0.02)	(0.03)	(0.03)
Income U	Asian	Hispanic	White	Race U
1.4	0.16	-0.21	0.69	0.18
(0.03)	(0.05)	(0.05)	(0.04)	(0.05)

Notes: Coefficient estimates of equations 8 via maximum likelihood. Standard errors in parenthesis and clustered by ZIP code. Dependent variable is an indicator that takes the value of one for customers who adopt an online grocery platform during the sample period. ZIP code fixed effects displayed in Figure 6 panel (c). Estimates use platform adoption choice of 1,512,981 customers. The log-likelihood is $-138,321$, McFadden's Pseudo R^2 is 0.10, and the likelihood ratio test statistic (within) is 14,404 (df = 10).
Source: Author's calculations using 1,690 ZIP codes with at least 500 customers from the constant panel of customers and at least one customer who lives there and adopts a platform.

model frames the fundamental trade off that determines adoption as one between the value of online grocery platforms versus trips to offline stores.[18]

Are the average welfare gains small or large? At the end of my sample period, a small share of consumers had ever tried, much less adopted, an online grocery platform, such that population-wide welfare effects are necessarily small. However, compared with the welfare estimates in Dolfen et al. (2023) for total online retail growth through 2017, the gains from the early online grocery delivery market are of similar magnitude.

4.E The Effects of a Mature Online Grocery Market

To study how the transition to a more mature online grocery market reshapes consumer welfare and store visits, I simulate a counterfactual increase in platform adoption. The low adoption rates during my sample period reflect slow adoption typical in new product markets and early platform quality issues, rather than a large-scale rejection by consumers. The COVID-19 pandemic accel-

[18]The model does not by itself decompose platform value into possible mechanisms, such as convenience, price, and variety. To evaluate possible price and variety gains, appendix section 9.A maps the reduced-form pre- and post-adoption spending changes into welfare using a CES benchmark. Adopting an online grocery platform in that framework is equivalent to a 3.1% decline in the price of online retail and a welfare gain of 0.9% of consumption for adopters. Given low adoption in the sample, population-wide gains from these mechanisms are small, consistent with most gains operating through the time savings gained by switching from offline stores to online grocery platforms.

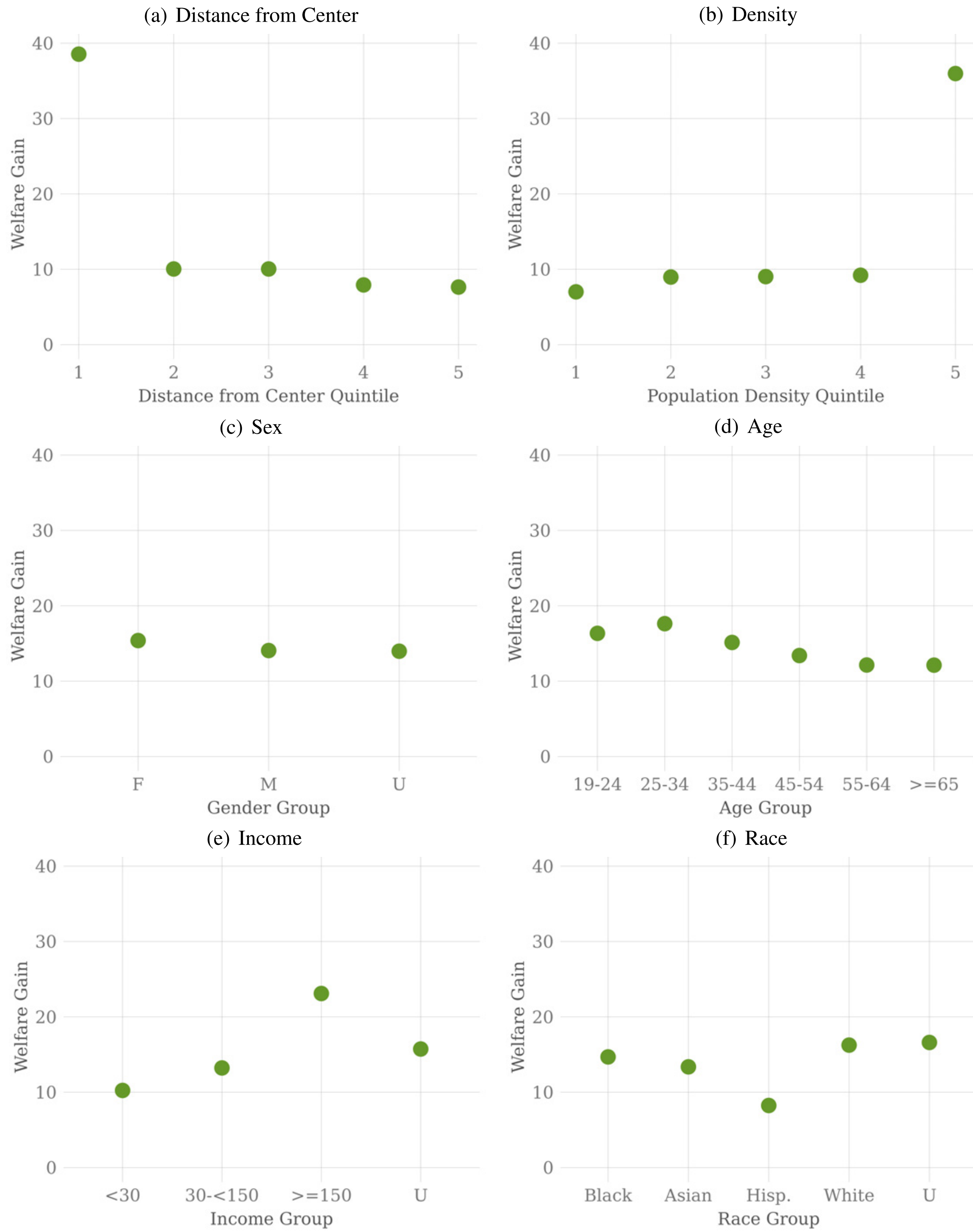
erated the arrival of the more mature market (Relihan et al. 2020). A 2022 USDA survey shows that 19.3% of customers shopped for groceries online in the last month, half primarily through online delivery (Restrepo and Zeballos 2024). The model estimated here can estimate the differential impact of this dramatic expansion for consumers and brick-and-mortar stores, apart from a retail supply response or concurrent pandemic-related impacts. To model this counterfactual in a straightforward and easily interpretable way, I target a population adoption rate of 10% through increasing the platform value by 120% for each ZIP code in equation 10. This maintains relative platform quality across space. I then study variation in consumer welfare and store visits across different markets and consumer groups.

Figure 7 shows the distribution of welfare gains to consumers from the continued expansion of online grocery platforms. Whether measured by distance to the CBD or population density (panels (a) and (b)), the welfare gains in the most urban ZIP codes are roughly four times larger than those for other ZIP codes. These geographic disparities far outweigh those between demographic groups in panels (c) - (f). Among these socioeconomics, welfare gains are more disparate by income, with the highest income consumers gaining twice that of the lowest income consumers. Differences are smaller across other categories, with larger gains for young, female, and White consumers.

Table 3 shows how more widespread adoption affects trip choices across markets and consumers. Aggregated across trip types, average visits to grocery stores decline by 0.7% and increase to restaurants and grocery stores each by 0.1%. However, the first two rows of Table 3 show that changes in trip patterns away from trips including the grocery store to other trip types are most pronounced in urban markets. The largest decline is for the trip to the grocery store chained with the general goods store, which falls in frequency by 1.0% on average, with distance and population elasticities of 0.4 and -0.2, respectively. The largest increase is for the restaurant chained with the general goods store, which is 1.1% more frequent on average, with distance and density gradients of -0.6 and 0.2, respectively. The restaurant alone trip shows steeper distance and density gradients than the general goods store alone, meaning new restaurant visits are the most concentrated in dense, central markets.

Table 3 also shows pronounced heterogeneity in trip changes across consumer socioeconomics. More widespread online grocery platform adoption shifts the customer base of in-person grocery

Figure 7: Effects of Mature Online Grocery Platform Market on Consumers



Notes: Welfare gains estimated via equation 11 for consumers under counterfactual platform values in today's mature online grocery platform delivery market.

Source: Author's calculations using data for 1,394,570 customers. Customers are from the balanced panel, have a complete set of representative "preferred stores", and reside in a ZIP code for which platform value is estimated and measures of distance from city center and population density are available.

Table 3: Effects of Mature Online Grocery Platform Market on Trips

Dependent Variable: Ln Trip Change							
Trip grn =	001	010	011	100	101	110	111
Ln Dist.	-0.14	-0.25	-0.57	0.33	0.53	0.44	0.24
	(0.07)	(0.13)	(0.29)	(0.17)	(0.27)	(0.23)	(0.13)
Ln Density	0.04	0.09	0.20	-0.14	-0.19	-0.17	-0.09
	(0.03)	(0.06)	(0.12)	(0.07)	(0.11)	(0.09)	(0.05)
Ln Age	0.04	0.07	0.16	-0.04	-0.08	-0.04	-0.01
	(0.04)	(0.07)	(0.18)	(0.12)	(0.18)	(0.17)	(0.09)
Sex F	-0.04	-0.05	0.01	-0.04	0.00	-0.04	-0.04
	(0.01)	(0.01)	(0.04)	(0.03)	(0.03)	(0.03)	(0.02)
Sex U	-0.02	-0.04	-0.06	0.04	0.07	0.06	0.03
	(0.02)	(0.04)	(0.08)	(0.05)	(0.08)	(0.07)	(0.04)
Income 2	0.07	0.07	0.08	-0.01	-0.09	-0.06	-0.04
	(0.05)	(0.08)	(0.17)	(0.09)	(0.16)	(0.14)	(0.08)
Income 3	0.15	0.17	0.26	-0.06	-0.29	-0.17	-0.08
	(0.11)	(0.17)	(0.39)	(0.20)	(0.38)	(0.30)	(0.16)
Income U	0.10	0.11	0.12	0.00	-0.11	-0.05	-0.03
	(0.07)	(0.10)	(0.22)	(0.11)	(0.20)	(0.16)	(0.09)
Asian	0.18	0.28	0.42	-0.19	-0.38	-0.25	-0.15
	(0.08)	(0.12)	(0.22)	(0.12)	(0.20)	(0.16)	(0.09)
Hispanic	0.13	0.18	0.24	-0.11	-0.23	-0.14	-0.10
	(0.05)	(0.07)	(0.11)	(0.06)	(0.11)	(0.08)	(0.05)
White	0.22	0.34	0.53	-0.26	-0.50	-0.33	-0.19
	(0.11)	(0.18)	(0.36)	(0.20)	(0.34)	(0.26)	(0.15)
Race U	0.12	0.21	0.47	-0.28	-0.45	-0.37	-0.21
	(0.07)	(0.13)	(0.30)	(0.18)	(0.29)	(0.25)	(0.14)
Mean	0.26	0.47	1.13	-0.61	-1.00	-0.83	-0.45
Adjusted R^2	0.35	0.38	0.38	0.39	0.38	0.38	0.38

Notes: Estimates for $LnTripChange_{iz} = \mathbf{x}'_{zi}\boldsymbol{\beta} + \phi_c$. The vector $\mathbf{x}_{zi}$ captures the distance from city center and population density of the customer's residential ZIP code, along with their socioeconomic characteristics of age, sex, income, and race where z indexes ZIP codes and i indexes customers. The ϕ_c are CBSA fixed effects. Clustered (CBSA) standard errors in parentheses. Table D3 shows changes in visits for grocery, restaurant, and general goods aggregated across trip types.
Source: Author's calculations using data for 1,394,570 customers. Customers are from the balanced panel, have a complete set of representative "preferred stores", and reside in a ZIP code for which platform value is estimated and measures of distance from city center and population density are available.

stores toward lower-income and non-White consumers, while that for restaurants and general goods stores is more male, high-income, and White. The estimates show that older and high-income consumers reallocate more away from trips that include a grocery store toward trips without a grocery store, consistent with greater scope to reallocate time to new trips. In contrast, female consumers exhibit smaller increases in restaurant alone and general goods alone trips and substitute more to the none trip, consistent with reallocating time savings to non-market activities. By race, Black and White consumers bracket the degree of redistribution, with Black consumers reallocating the least away from trips that include a grocery store and White consumers reallocating the most.

In summary, the expansion of the online grocery delivery market brings disproportionate benefits to consumers in the most urban markets. While new online grocery delivery platforms only partially replace grocery store visits, how much grocery stores are hurt by this expansion depends on further platform adoption and whether platforms independently deliver groceries or operate in partnership with local stores. Spillover effects through time use and trip chaining can create further winning and losing stores from this expansion. Gains to services are also heavily concentrated in the most urban areas. Finally, as the market matures, general equilibrium effects will have stronger welfare implications through changes in prices, store and consumer locations, and access to store variety. For example, sharp increases in restaurant trip frequency in central urban neighborhoods would likely induce additional restaurant entry that further improves welfare in those locations through greater access to variety and reduced distance costs. Quantifying these general equilibrium responses and their welfare effects is a promising topic for future research.

5 Conclusion

This study sheds light on the transformation of local offline economics in response to the continued rise of online retail. The results reject the view that online retail constitutes a wholesale "retail apocalypse" for cities, but rather furthers a transformation of cities toward places that provide value through consumer services. Finding that high-income, urban consumers use additional time from online grocery delivery to purchase more services is also in line with decades long trends in allocating free time toward leisure activities (Aguiar and Hurst 2007b; Su 2022b). These results are particularly relevant for the effects of the pandemic-driven acceleration of the online grocery market.

The expansion of online retail has influence beyond retail markets, with particular relevance to the broader dynamics of labor and commercial real estate markets. As of December 2022, nearly 15 million people were employed in brick-and-mortar retail jobs, roughly 11% of private sector employment.[19] This research indicates a reshuffling of workers from goods retail to service employment concentrated in urban areas. In addition, changing spatial consumption patterns will differentially affect the value of commercial property. The results presented here imply higher values near residential neighborhoods and retail dense locations, especially those anchored by services. Local governments may struggle to meet their funding needs if the revenue from traditional sales taxes on goods decline more than those on services rise.[20] Governments could respond by restructuring their tax code while liberalizing land use restrictions to encourage service-dense retail areas closer to residences.

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Supplemental Appendix

6 Appendix: Data and Sample

6.A Raw Datasets

The primary data are accessed through the JPMorgan Chase Institute and include transactions made by tens of millions of anonymized Chase customers starting in October 2012. Each transaction record contains information used to determine the payment amount, date, location, channel, and merchant. Customers are anonymized and linked across accounts using unique identifiers.

Customers: I refer to individual customers rather than households due to the ambiguous nature of the latter in the dataset. Customers sharing an address may suggest a household, yet a comprehensive household definition would require each member to be a Chase customer. Moreover, certain accounts may represent expenditures by multiple individuals, such as joint account holders. For these accounts, transactions and demographic information are tied to the primary account holder. Socioeconomic data for each customer are sourced as follows:

1. Sex is imputed from names.

2. Only customers at least 18 years old are included.

3. Annual income estimates are derived from deposit account flows and are recorded as unknown when annual inflows are derived from fewer than 10 separate months of inflows, annual inflows are less than twelve thousand dollars, or customers exclusively hold credit card accounts.

4. Modeled race data was obtained in 2021 from a third party for the JPMorganChase Institute to conduct economic research examining financial outcomes by race. The data was matched to internal banking records using encrypted quasi-identifiers. This de-identified file that contains banking records and modeled race is only available to the JPMorganChase Institute. White includes White and races other than Black, Asian, and Hispanic. Unknown race refers to customers for whom no modeled race is available.

Locations: Customer and merchant ZIP codes are based on mailing addresses and card terminal locations, respectively.

Payment Channel: I classify a transaction as online if the card was present at the time of purchase, the merchant is a known, large online-only retailer, or the store location is characterized by a website or phone number.

Digital wallet payments (e.g., Apple Pay) are classified as card-present transactions when made at a physical terminal, as with physical card payments. Payments made through a merchant app or website that are processed with an online terminal are classified as card-not-present transactions, even if the consumption occurs in person.

Product: Each transaction carries a four-digit merchant category code (MCC) that characterizes the goods or services sold. MCCs are standardized across the financial industry for retail transactions. For traditional brick-and-mortar retail merchants, these appear to consistently reflect the predominant business of the merchant. For example, Walmart transactions are typically classified under the MCC for grocery stores and supermarkets. Merchants with multiple lines of business (e.g., grocery stores with pharmacies or gas stations) typically use separate terminals programmed with different MCCs to capture these distinctions. However, for emerging spending types, I find the MCC alone are inadequate for classification of some products. For example, many online grocery and restaurant transactions for new online-only merchants are coded as miscellaneous services instead of grocery or restaurant. To address these issues, I define a broader "product" categorization from MCCs, supplemented by manual mapping of leading merchants for specific categories like online groceries and restaurants.

Geographic and economic context: To complement the transactions data, I use geographic and demographic data from the Census and Google Maps.

1. ZIP code centroids, population density, and their correspondence to cities are sourced from Census Gazetteer files. Cities are defined as Core Based Statistical Areas.

2. To calculate the distance of each ZIP code from its city center, I follow the methodology of Holian (2019) and Couture and Handbury (2020), using the location of the earliest recorded city hall in each city as the city center. Geographic coordinates for these locations are obtained from Google Maps. In cases where the historical city hall location is unavailable, I use alternative municipal buildings, such as the historical courthouse or the oldest municipal building in the city. For example, Miami's city hall was moved to its current location outside

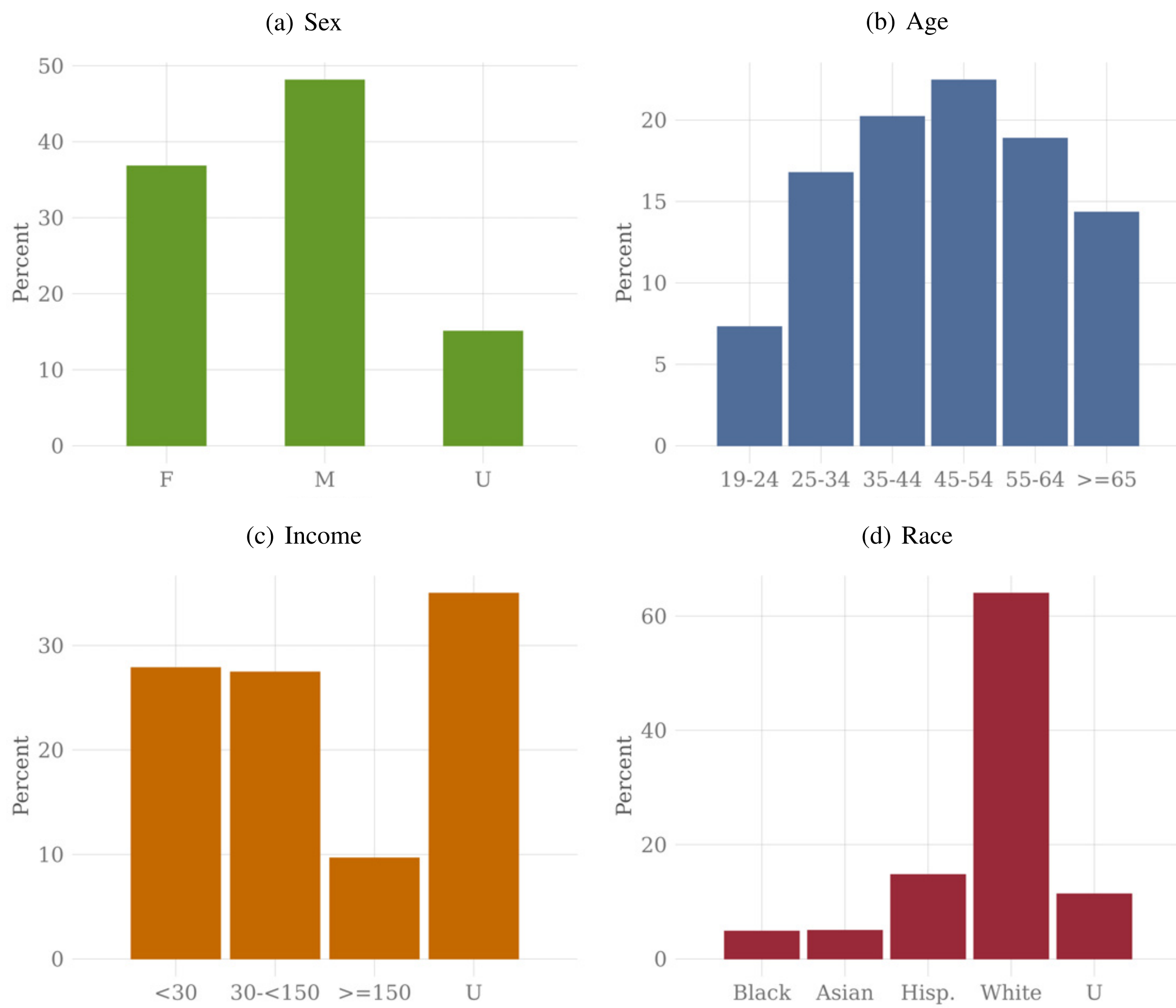
downtown in 1954, so I use the location of the earlier city hall as the city center. I define the ZIP code distance from city center as the distance from this point to the centroid of the ZIP code.

3. Demographic data for each ZIP code are sourced from the 2015-2019 American Community Survey.

6.B Customer Panel

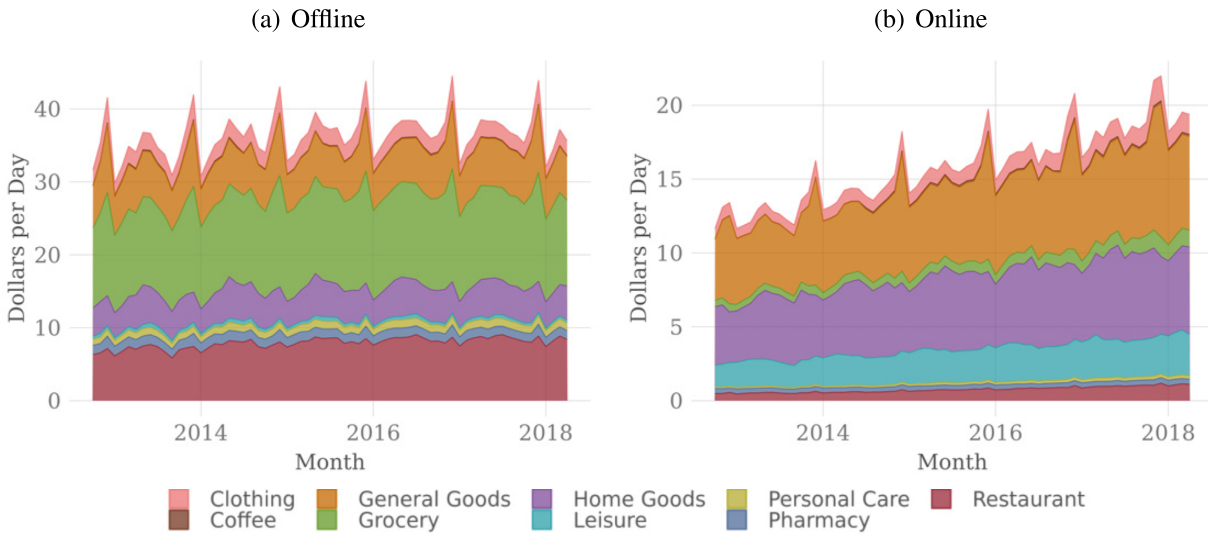
I construct a balanced panel of 7.2 million customers who make at least three transactions every month from October 2012 through April 2018. Restricting the sample in this way excludes customers who are new to the bank or those who stop using their Chase cards, focusing instead on those with a consistent usage pattern. Figure A1 provides the socioeconomic profile for customers in the balanced sample. Ganong and Noel (2019) use similar restrictions to build a sample from the same data source and show extensive benchmarking to external sources to show its representativeness. Importantly, they also show that Chase data have an absolute advantage to publicly available data for coverage of nondurable spending.

Figure A1: Panel Demographics



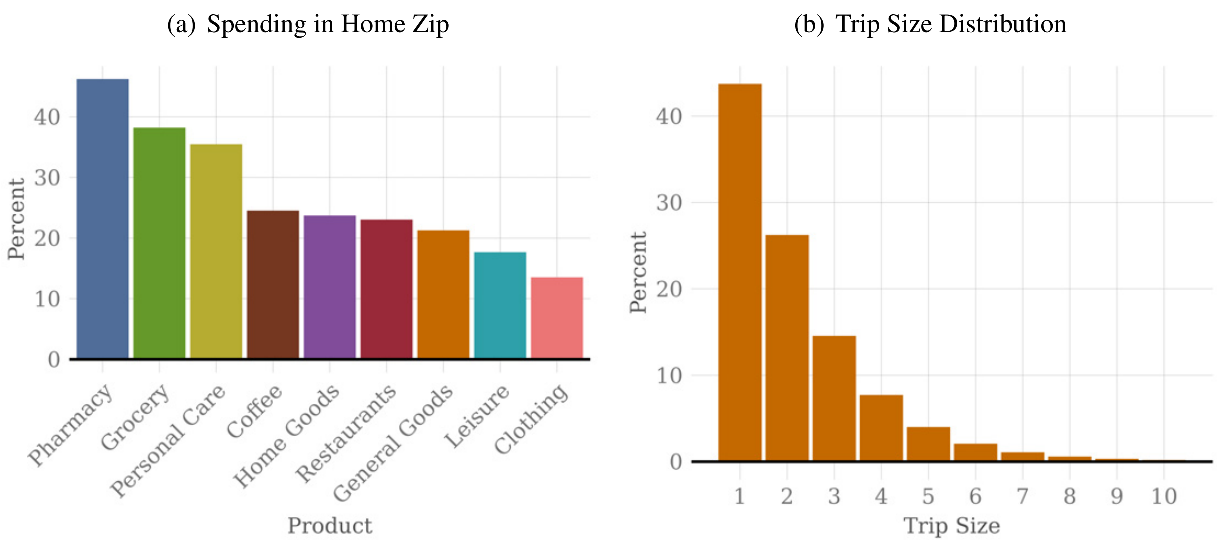
Notes: Gender, age, income, and race distributions in the customer panel. Uncategorized customers in a category are labeled with a "U".
Source: Author's calculations using the card transactions from a balanced panel of 7.2 million customers who make at least 3 transactions each month from October 2012 - April 2018.

Figure A2: Spending by Product and Channel



Notes: Average offline and online spending for everyday products in the customer panel.
Source: Author's calculations using the card transactions from a balanced panel of 7.2 million customers who make at least 3 transactions each month from October 2012 - April 2018.

Figure A3: Offline Trip Features



Notes: Panel (a) shows the share of everyday products purchased in customers' home ZIP codes. Panel (b) shows the distribution of the number of offline purchases made made on a day (assumed to be one trip) with a least one offline purchase.
Source: Author's calculations using the card transactions from a balanced panel of 7.2 million customers who make at least 3 transactions each month from October 2012 - April 2018.

7 Appendix: Platform Entry

In this appendix, I describe and validate the algorithm I develop to statistically measure the entry of online grocery platforms into cities and ZIP codes using 98 billion transactions made by an unbalanced panel of 77 million customers from October 2012 through January 2020. I also provide further descriptive figures and tables for platform entry across cities and ZIP codes. Note that it is not possible to scrape historical platform availability from platform websites because only the query webpage for availability is archived, not the search results for each ZIP code.

7.A Measurement

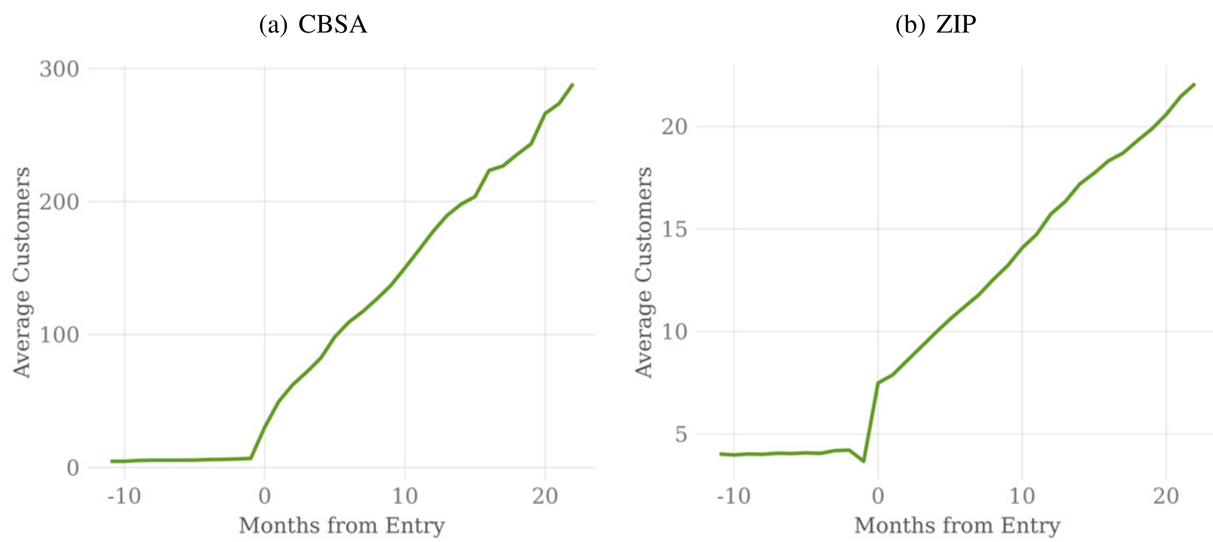
As a starting point, using the first transaction as the platform entry date for a ZIP code is ineffective due to address update delays and deliveries to other addresses. For example, if a consumer moves from ZIP code A (where a platform is unavailable) to ZIP code B (where it is available) but doesn't update their address, transactions may appear from ZIP code A. To address this, I estimate the actual entry month by identifying structural breaks in the platform use time series for each ZIP code, following Hansen (2000). Specifically, I calculate the number of customers C_{zpm} in each month m using each platform p in each ZIP z and then run a series of regressions over all possible breakpoints λ^*_{zpm} that could be the entry month of the form:

$$C_{zpm} = \alpha_z + \beta_z \mathbf{1}[\lambda_{zpm} \geq \lambda^*_{zpm}] + \varepsilon_{zpm}. \tag{12}$$

To reduce the number of regressions and avoid selecting a platform use decline as the entry date, I limit possible breakpoints to months between the first observed use and peak use, and specifically to the month following the peak growth month. For example, if customer transactions start in June 2013 and peak in October 2015, I identify the largest growth month within this range, say April 2014, and use all months from June 2013 to May 2014 as candidate breakpoints. Customer growth in month m as $ln(C_{zpm}+5) - ln(C_{zpm-1}+5)$, adding five to penalize growth rates from very small customer counts before true entry. I then recover the t-stat from each regression and select the highest t-stat > 1.96 for each platform and ZIP code as the structural breakpoint. I disregard combinations where no t-stat > 1.96 exists. Additionally, I exclude platforms in ZIP codes that were active at the start of my sample, those with fewer than three months of activity or fewer than three customers on average, and ZIP codes with fewer than 100 customers at the start of my sample.

In practice, this process involves running hundreds of thousands of regressions to identify 15,748 distinct entries of grocery delivery platforms into ZIP codes. I also execute an equivalent exercise to measure 908 distinct entries of platforms into cities, with larger cities experiencing multiple entries and most small cities experiencing just one. Figure B1 shows the average number of platform users in a city or ZIP code before and after the platform entry, as defined by my algorithm. At entry, there is a substantial and distinct jump in the average number of recorded platform users.

Figure B1: Platform Use Around Platform Entry



Notes: Panel (a) shows the average number of customers using a platform in a city before and after the month of entry into that city. Panel (b) shows the equivalent for ZIP codes.
Source: Author's calculations using 98 billion transactions made by an unbalanced panel of 77 million customers from October 2012 through January 2020. Includes entry of 17 online grocery delivery platforms.

I also externally validate this entry algorithm using public news announcement of platforms' entries into cities. Local and sometimes national news outlets often covered the entry of these new online grocery platforms entries into cities, mostly for large and nationally expanding platforms that are not wholly owned by an established grocery store chain. I searched news archives for any reference to a platform's entry dates into cities broadly as well as targeted searches for entry into the top 20 largest cities by 2010 population. Examples of these articles are Instacart (2014) and

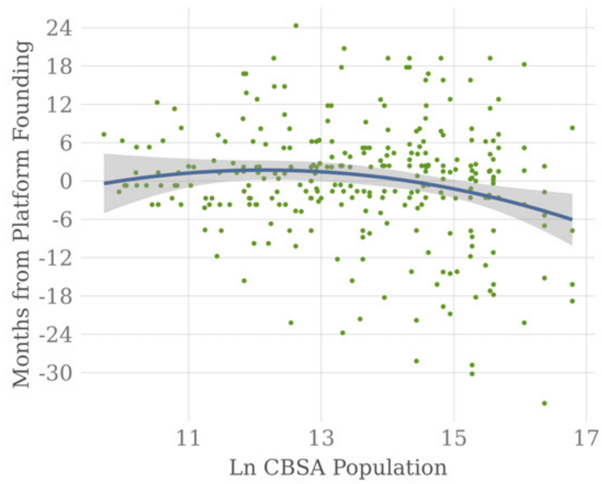
Tuohy (2017). A similar exercise for ZIP codes is precluded by the fact that only a couple of news articles mention specific neighborhoods in which platforms are initially made available.

This exercise reveals important context for measuring entry with transactions data. First, small initial entries into a few ZIP codes relative to a much larger later surge in availability elsewhere in the city can sometimes lead the algorithm to choose the latter date as the initial city entry. These entries are too late at the city level. Second, discrepancies between entry measured in the transaction data and entry covered in the news rise after 2017. This is driven by too early detection of entry in the transaction data for platforms that enter in 2018 or later. I therefore restrict the study of early adopters for those where initial ZIP entry occurs by January 2017. Finally, of course, some entries covered in the news are undetected in the transactions data, either because the Chase customer base in a city is not big enough and/or the platform was not popular enough in the city to create a statistically significant surge in customer use at entry.

In the end, I find 49 platform entries into cities that are both measured in the transactions data from April 2013 to January 2017 and covered in the news. Among these, 30 (47%) agree exactly on the month of entry. Another seven (cumulatively 61%) are measured in the transactions data with a one-month delay relative to news coverage due to entry that occurs late in a month. 86% are within one a year of the entry covered in the news.

7.B Descriptive Figures and Tables

Figure B2: Early Platform Entry into Larger Cities

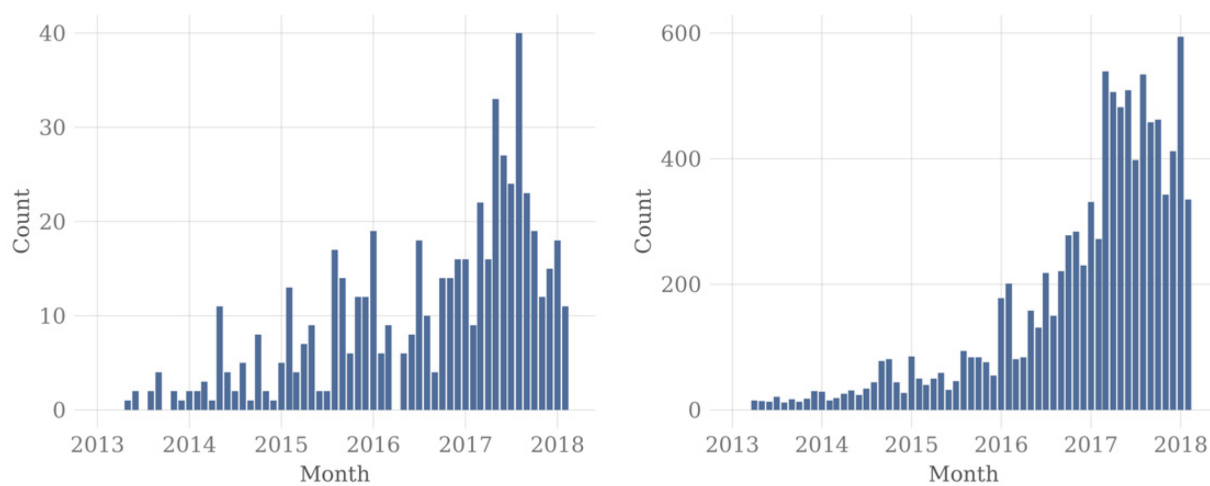


Notes: Normalized number of months between the founding month of the platform and its entry into a new CBSA correlated with CBSA population. The months from founding are normalized by the average months from founding for each platform to account for platform differences in speed of expansion and time of founding relative to the sample period. The line is a quadratic fit for the correlation.
Source: Author's calculations using 98 billion transactions made by an unbalanced panel of 77 million customers from October 2012 through January 2020. Includes entry of 17 online grocery delivery platforms.

Figure B3: Platform Entry over Time

(a) CBSA Entries

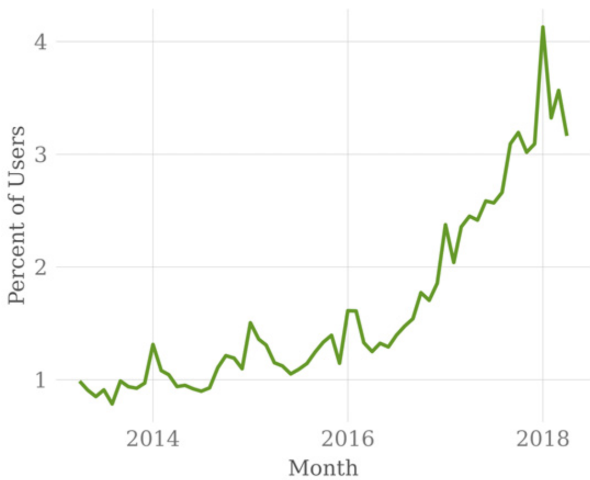
(b) ZIP Entries



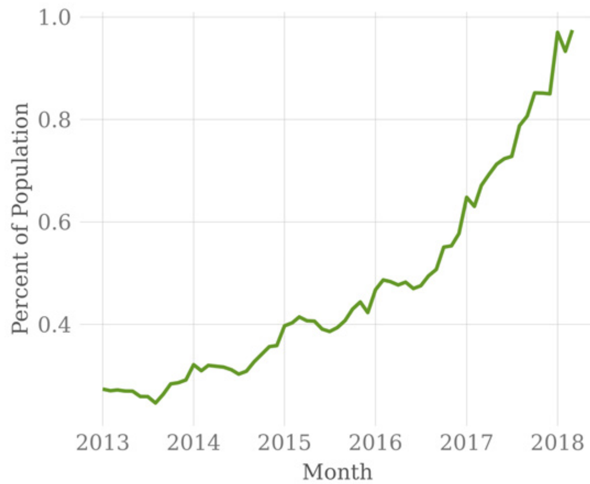
Notes: Panel (a) shows the number of city entries of online grocery platforms each month. Panel (b) shows the equivalent for ZIP codes.
Source: Author's calculations using 98 billion transactions made by an unbalanced panel of 77 million customers from October 2012 through January 2020. Includes entry of 17 online grocery delivery platforms.

Figure B4: Grocery Platform Use

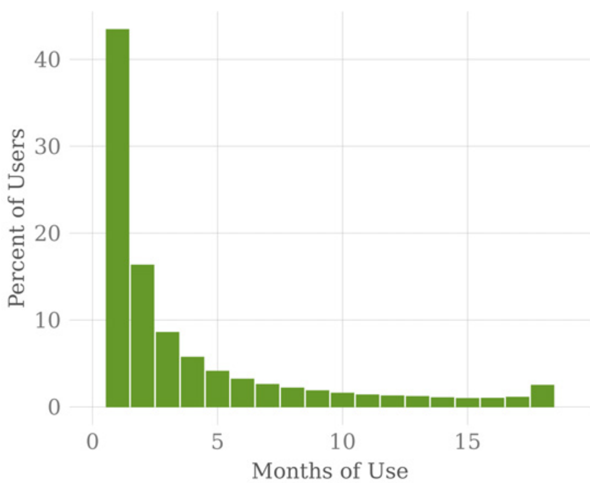
(a) Initial Use over Time



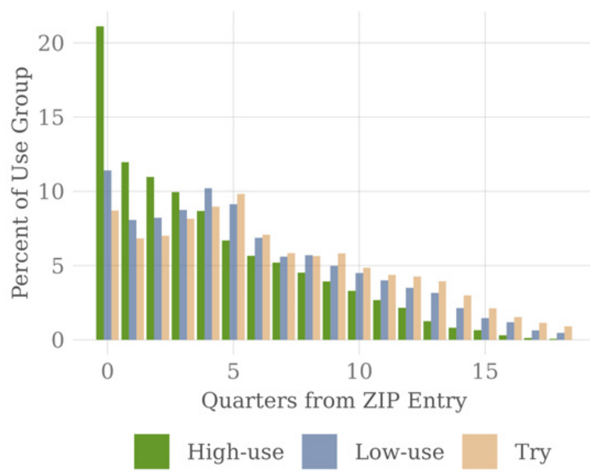
(b) Cumulative Initial Use over Time



(c) Use Strength



(d) Use Speed



Notes: Panel (a) shows the percent of customers who initially use a platform each month as a percent of all platform users. Panel (b) shows the percent of customers who have ever tried a platform over the sample period. Panel (c) shows the distribution of the number of months of platform use among platform users. Panel (d) shows the distribution of percent of users who first try a platform each quarter from the platform's entry into their ZIP code. The distribution is divided by platform use groups, where high-use is defined as using a platform in at least five months of the first 18 months after initial use, low-use is defined as using a platform two to four months of the first 18 months after initial use, and try is defined as a single month of use.

Source: Author's calculations using the card transactions from a balanced panel of 7.2 million customers who make at least 3 transactions each month from October 2012 - April 2018.

Table B1: Characteristics of ZIP Entry

(a) Months from CBSA Entry

	(1)	(2)	(3)
Ln Distance from Center	1.44	1.41	1.73
	(0.41)	(0.38)	(0.25)
ln Density	-2.63	-1.91	-2.95
	(0.35)	(0.18)	(0.27)
Non-white Share			0.010
			(0.009)
Ln Median Income			-8.62
			(0.78)
CBSA FE	Y	Y	Y
Platform FE	N	Y	Y
Observations	14,472	14,472	14,472
Adjusted R2	0.32	0.51	0.54

(b) Total Entry

	N. Entries		Ever Entry	
	(1)	(2)	(3)	(4)
Ln Distance from Center	-0.04	-0.07	-0.02	-0.04
	(0.03)	(0.02)	(0.01)	(0.01)
ln Density	0.30	0.29	0.14	0.13
	(0.02)	(0.02)	(0.01)	(0.01)
Non-white Share		0.003		0.002
		(0.001)		(0.0003)
Ln Median Income		0.77		0.35
		(0.05)		(0.02)
CBSA FE	Y	Y	Y	Y
Observations	15,198	15,198	15,198	15,198
Adjusted R2	0.60	0.65	0.50	0.55

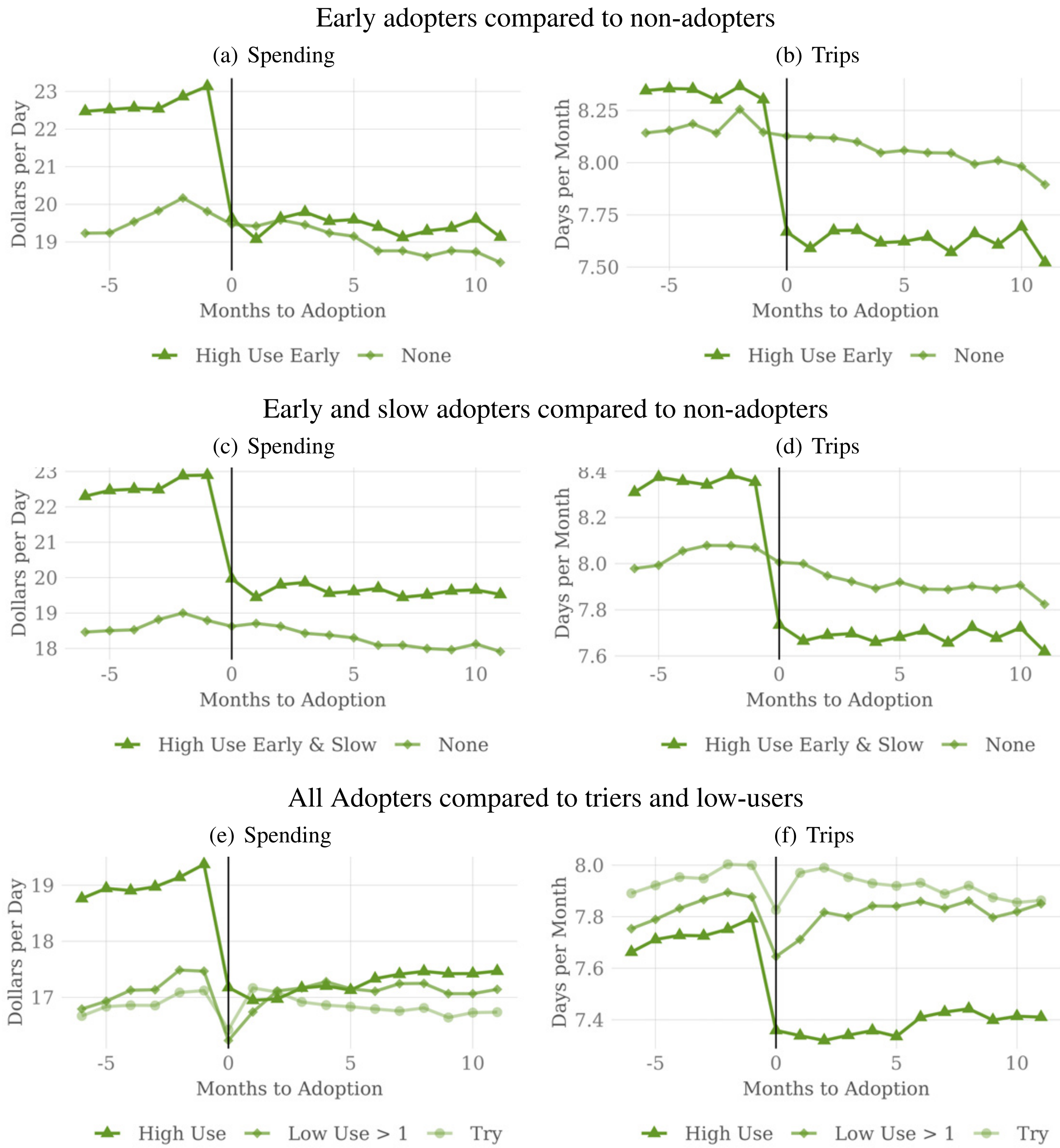
Notes: Panel (a) estimates $Months_z = \beta_1 LnDist_z + \beta_2 LnDensity_z + \beta_3 NonWhiteShare_z + \beta_4 LnMedianIncome_z + \phi_c + \phi_p$ where z indexes ZIP codes and the ϕ are CBSA and platform fixed effects. Mean months from CBSA Entry is 15.3. Standard errors are clustered by platform and city. Panel (b) shows estimates for total platform entries or whether a ZIP code ever experienced a platform entry using a similar specification without platform fixed effects. Mean number of entries is 1.0 and mean of ever entry is 0.5. Standard errors are clustered by CBSA.
Source: Author's calculations using 98 billion transactions made by an unbalanced panel of 77 million customers from October 2012 through January 2020. Includes entry of 17 online grocery delivery platforms.

8 Appendix: Reduced Form

8.A Empirical design

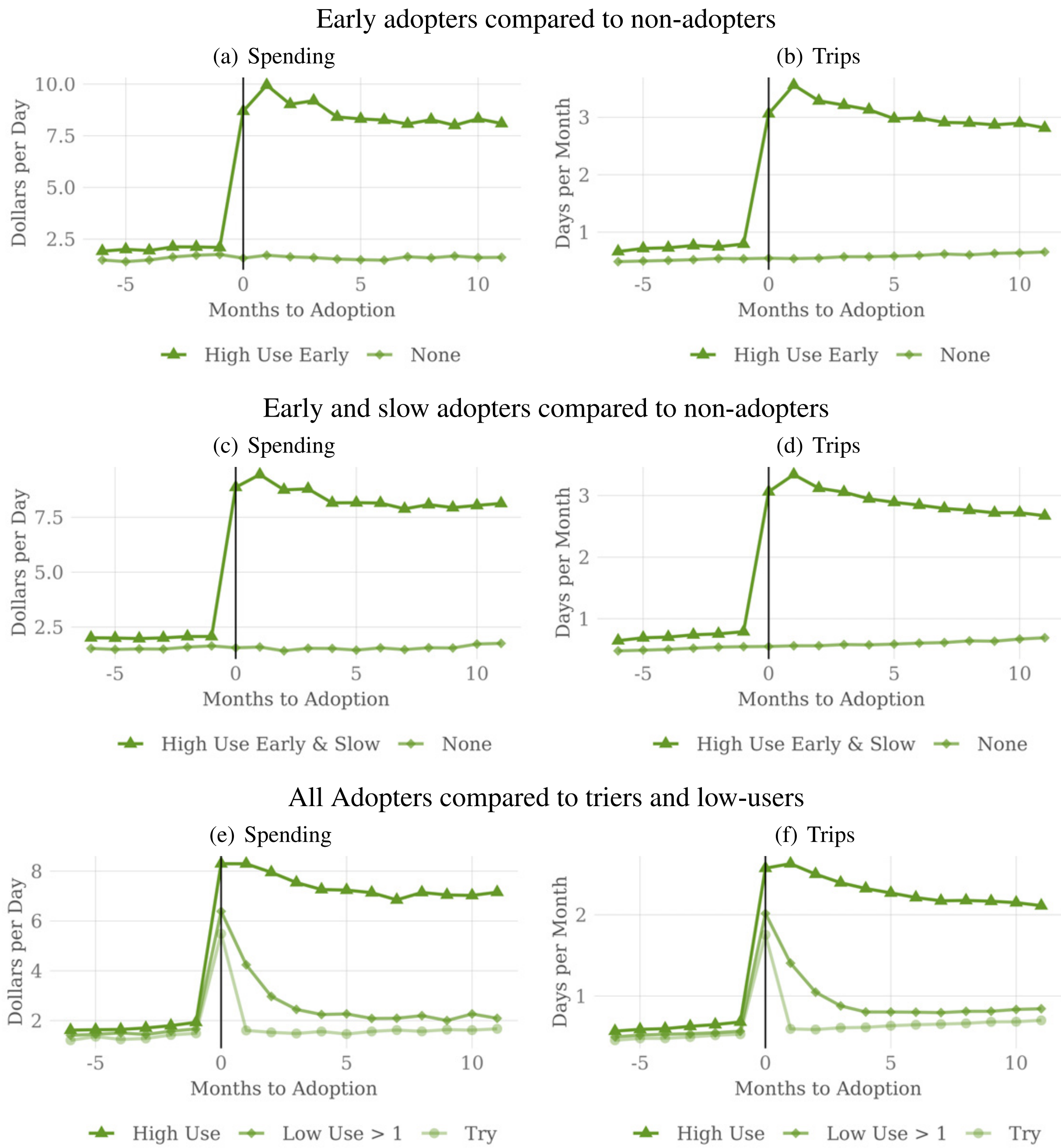
In this appendix, I provide further detail on matched treatment and control groups. Table C1 shows coefficients and standard errors for the socioeconomics and spending and trip patterns that are utilized in matching early adopters who adopt platforms in September 2015. Aggregated across all matches, balancing tests show that the matched controls closely parallel the consumption trends of everyday products for early adopters prior to adoption. Table C2 shows substantial improvement in two-sided t-stats after matching for the pre-adoption covariates used in matching. Reassuringly, Table C3 also shows similar improvements on those not used in matching. The top panels of Figures C1 and C2 further emphasize the parallel month-by-month trends in average spending and trips for offline and online groceries. The trends in averages are similarly parallel for other products, but I exclude these figures for brevity. This is consistent with the assumption that adoption at entry is uncorrelated with unobserved demand shocks.

Figure C1: Offline Grocery Spending and Trips for Alternative Treatment and Control groups



Notes: Trends around adoption in average spending and trips for offline grocery for the preferred treatment and control group (panels (a) and (b)) and two alternative approaches. See text for details.
Source: Author's calculations using the transactions of 11,418 early adopters of online grocery platforms and their matched controls from a set of platform non-users in panels (a) and (b). Panels (c) and (d) include 22,541 early and slow adopters and their matched controls from a set of platform non-users. Panel (e) and (f) include 55,831 adopters (regardless of adoption timing relative to entry) and their matched controls among platform triers and low-users.

Figure C2: Online Grocery Spending and Trips for Alternative Treatment and Control Groups



Notes: Trends around adoption in average spending and trips for online grocery for the preferred treatment and control group (panels (a) and (b)) and two alternative approaches. See text for details.
Source: Author's calculations using the transactions of 11,418 early adopters of online grocery platforms and their matched controls from a set of platform non-users in panels (a) and (b). Panels (c) and (d) include 22,541 early and slow adopters and their matched controls from a set of platform non-users. Panel (e) and (f) include 55,831 adopters (regardless of adoption timing relative to entry) and their matched controls among platform triers and low-users.

Table C1: Early Adopter Predictors in September 2015

Category	Covariate	Level		6MΔ	
Sex	Female	0.448	(0.177)		
	Unknown	0.410	(0.235)		
Age Bins	Age 2	2.327	(0.646)		
	Age 3	1.729	(0.654)		
	Age 4	0.760	(0.670)		
	Age 5	0.022	(0.714)		
	Age 6	-1.957	(1.113)		
Income Bins	Income 2	0.451	(0.303)		
	Income 3	0.345	(0.361)		
	Unknown	1.319	(0.289)		
Race	Asian	-1.268	(0.593)		
	Hispanic	-0.970	(0.547)		
	White	-0.160	(0.386)		
	Unknown	-0.247	(0.451)		
Offline Spending	Restaurant	-0.012	(0.010)	0.011	(0.037)
	Clothing	0.014	(0.014)	0.016	(0.046)
	Grocery	0.009	(0.003)	-0.003	(0.029)
	General	0.007	(0.003)	-0.013	(0.021)
	Home	0.001	(0.002)	0.010	(0.011)
	Leisure	-0.034	(0.043)	0.122	(0.091)
	Pharmacy	0.076	(0.037)	-0.056	(0.097)
	Personal	0.029	(0.028)	-0.055	(0.114)
	Coffee	0.019	(0.132)	0.074	(0.744)
Offline Trips	Restaurant	0.032	(0.018)	0.022	(0.092)
	Clothing	-0.360	(0.103)	0.237	(0.315)
	Grocery	-0.012	(0.019)	0.097	(0.116)
	General	-0.069	(0.045)	0.360	(0.176)
	Home	-0.032	(0.050)	-0.389	(0.184)
	Leisure	0.115	(0.097)	-0.340	(0.331)
	Pharmacy	-0.039	(0.057)	0.403	(0.225)
	Personal	0.203	(0.087)	0.302	(0.372)
	Coffee	0.034	(0.046)	0.320	(0.286)
Online Spending	Restaurant	-0.009	(0.013)	0.024	(0.035)
	Clothing	-0.057	(0.023)	0.127	(0.050)
	General Goods	-0.013	(0.006)	-0.004	(0.013)
	Home Goods	-0.003	(0.003)	-0.003	(0.009)
Online Trips	Restaurant	0.182	(0.038)	-0.526	(0.197)
	Clothing	0.527	(0.115)	-0.928	(0.334)
	General Goods	0.205	(0.027)	-0.115	(0.128)
	Home Goods	0.110	(0.091)	0.607	(0.334)
Travel Spending	Fuel	0.032	(0.027)	-0.103	(0.076)
	Transportation	0.075	(0.038)	0.283	(0.096)
Travel Trips	Fuel	-0.005	(0.014)	0.201	(0.155)
	Transportation	-0.082	(0.045)	-0.143	(0.165)
Observations	39,136				
Log Likelihood	-8,801				

Notes: Estimated coefficients from a logit regression of early platform adoption on socioeconomic characteristics and levels and changes in spending and trips using early adopters who adopted a platform in September 2015 and a random sample of platform non-adopters from their same ZIP codes. Levels are the 3-month average prior to adoption and changes are the average month-over-month change in the six months prior to adoption. Matching is with replacement and within ZIP code. Standard errors in parentheses.
Source: Author's calculations using the transactions of 39,136 early adopters of online grocery platforms and potential controls from a set of platform non-users.

Table C2: Balance on Covariates Used in Matching

		Means			*t-stats*	
Category	Covariate	Adopters	Non-adopters		Non-adopters	
			(All)	(Matched)	(All)	(Matched)
Socioeconomics	Female	0.42	0.36	0.42	9.41	-0.44
	Sex U	0.15	0.18	0.14	-6.56	-0.22
	Age Group	2.06	2.62	2.05	-37.1	0.86
	Income Group	2.86	2.45	2.99	29.39	-4.28
	Race Group	3.83	3.65	3.85	17.6	-1.03
Offline Spending 3M	Restaurant	17.41	8.97	16.79	35.39	2.56
	Clothing	5.46	2.69	5.23	21.06	1.34
	Grocery	22.53	12.69	19.94	38.29	5.85
	General	10.51	6.26	9.45	20.85	2.8
	Home	7.40	4.41	7.68	11.61	-0.20
	Leisure	0.89	0.52	0.90	10.02	0.24
	Pharmacy	2.47	1.41	2.26	24.11	3.20
	Personal	3.16	1.18	3.02	33.97	2.24
	Coffee	0.40	0.16	0.36	21.71	3.23
Offline Trips 3M	Restaurant	11.73	7.94	11.78	40.13	0.09
	Clothing	1.17	0.84	1.14	20.42	1.18
	Grocery	8.29	7.02	8.18	19.13	1.35
	General	3.28	2.67	3.18	18.71	1.32
	Home	1.42	1.14	1.43	11.50	-0.28
	Leisure	0.50	0.37	0.52	12.47	-0.46
	Pharmacy	2.13	1.49	2.03	23.03	2.41
	Personal	1.34	0.66	1.34	36.45	0.50
	Coffee	1.17	0.49	1.11	22.29	2.02
Offline Spending 6MΔ	Restaurant	0.03	0.00	0.06	0.72	-0.29
	Clothing	0.09	0.01	0.02	2.05	0.92
	Grocery	0.11	0.04	0.12	1.90	0.26
	General	0.19	0.07	0.05	2.11	0.39
	Home	0.00	-0.03	-0.03	0.34	-0.46
	Leisure	0.01	0.00	0.04	1.03	-0.56
	Pharmacy	0.01	0.00	0.00	0.79	-0.19
	Personal	0.00	0.00	-0.01	-0.33	-0.43
	Coffee	0.00	0.00	0.00	-0.36	1.11
Offline Trips 6MΔ	Restaurant	-0.02	-0.02	0.00	0.09	-0.19
	Clothing	0.01	0.00	0.01	2.03	0.28
	Grocery	-0.02	0.00	0.00	-1.41	-0.57
	General	0.02	0.01	0.02	1.33	-0.15
	Home	0.00	-0.01	0.00	0.58	-1.38
	Leisure	0.00	0.00	0.01	2.25	-0.80
	Pharmacy	0.00	0.00	0.00	-0.01	-0.32
	Personal	0.00	0.00	0.00	1.05	-0.14
	Coffee	0.00	0.00	0.00	0.16	1.11

Notes: Continued on following page.

Table C2: Balance on Covariates Used in Matching, continued

		Means			*t-stats*	
Category	Covariate	Adopters	Non-adopters		Non-adopters	
			(All)	(Matched)	(All)	(Matched)
Online Spending 3M	Restaurant	3.14	0.86	2.40	21.46	4.63
	Clothing	4.64	1.13	3.49	7.29	2.99
	General Goods	20.28	5.66	16.21	23.17	4.99
	Home Goods	15.95	5.23	13.51	6.31	1.28
Online Trips 3M	Restaurant	2.18	0.75	1.83	40.86	6.43
	Clothing	0.81	0.25	0.71	33.69	3.50
	General Goods	5.29	1.69	4.73	62.86	5.92
	Home Goods	0.87	0.35	0.80	25.76	2.20
Online Spending 6MΔ	Restaurant	0.10	0.01	0.06	1.78	0.95
	Clothing	0.04	0.02	0.08	0.18	0.35
	General Goods	0.74	0.09	0.47	5.33	1.58
	Home Goods	-0.16	0.02	-0.10	-0.97	-0.48
Online Trips 6MΔ	Restaurant	0.07	0.02	0.06	7.33	1.83
	Clothing	0.02	0.01	0.02	4.23	0.51
	General Goods	0.17	0.03	0.14	12.87	0.29
	Home Goods	0.01	0.00	0.00	0.93	0.94
Travel Spending 3M	Fuel	5.15	4.33	5.31	13.44	-1.83
	Transportation	2.89	1.55	2.54	24.66	5.19
Travel Trips 3M	Fuel	7.37	6.16	7.47	11.50	-0.25
	Transportation	2.81	1.44	2.28	17.26	4.95
Travel Spending 6MΔ	Fuel	-0.02	-0.03	-0.01	0.22	-0.52
	Transportation	0.07	0.01	0.05	3.05	0.67
Travel Trips 6MΔ	Fuel	-0.01	-0.01	0.01	0.46	-1.12
	Transportation	0.08	0.02	0.07	6.71	2.18

Notes: Continued from previous page. Mean differences in spending and trips in levels and changes between early adopters and non-adopters before and after matching. Variables in this table are included in the matching exercise. Levels are 3-month averages prior to adoption and changes are average month-to-month changes in the six months prior to adoption. Two-sided t-stats for the differences in means before and after matching are reported.
Source: Author's calculations using 3,805 early adopters of online grocery platforms, 2,226,224 non-adopters as potential matches, and 7,613 non-adopters matched as controls.

Table C3: Balance on Covariates Not Used in Matching

Category	Covariate	*Means* Adopters	Non-adopters (All)	Non-adopters (Matched)	*t-stats* Non-adopters (All)	Non-adopters (Matched)
Aggregate Spending 3M	Total spending	221.94	102.16	195.11	26.60	5.15
	Cash	0.07	0.07	0.09	0.10	-0.17
	Check	0.10	0.18	0.14	-2.81	-0.40
	Offline	99.07	56.24	92.98	37.52	4.45
	Online	122.99	45.97	102.22	20.15	4.71
Aggregate Trips 3M	Total spending	24.93	20.22	24.37	68.80	5.34
	Cash	0.00	0.01	0.01	-1.36	-1.10
	Check	0.00	0.00	0.00	-1.52	0.51
	Offline	20.51	17.33	20.55	33.55	0.20
	Online	16.03	8.22	14.07	93.95	14.50
	Single Distance	691.19	431.49	681.52	23.10	0.86
	Chained Distance	490.67	321.28	481.19	24.03	1.25
Aggregate Spending 6MΔ	Total spending	0.00	0.00	-0.01	0.63	1.13
	Cash	0.00	0.00	0.00	-0.06	-0.07
	Check	1.57	0.37	1.55	2.47	-0.59
	Offline	0.69	0.07	0.39	2.94	0.90
	Online	0.88	0.30	1.16	1.43	-1.12
Aggregate Trips 6MΔ	Total spending	0.00	0.00	0.00	-0.19	0.77
	Cash	0.00	0.00	0.00	0.26	0.28
	Check	0.07	0.01	0.07	5.47	1.15
	Offline	-0.01	-0.02	0.01	0.50	0.19
	Online	0.24	0.08	0.20	12.76	1.77
	Single Distance	8.46	5.80	7.92	1.30	0.38
	Chained Distance	5.63	3.79	5.66	1.38	0.18
Restaurant Spending 3M	Caterer	0.06	0.02	0.05	4.30	0.82
	Eating Place	13.46	6.79	13.00	31.75	2.41
	Drinking Place	0.42	0.22	0.52	9.80	-1.73
	Fast Food	3.47	1.94	3.23	32.75	3.33
Restaurant Trips 3M	Caterer	0.04	0.02	0.03	5.71	2.08
	Eating Place	7.14	4.44	7.16	39.41	0.74
	Drinking Place	0.24	0.15	0.28	10.91	-1.94
	Fast Food	6.42	4.34	6.42	28.38	-0.30
Restaurant Spending 6MΔ	Caterer	0.00	0.00	0.00	-0.91	0.24
	Eating Place	0.01	0.00	0.06	0.29	-0.59
	Drinking Place	0.01	0.00	0.01	1.19	-0.15
	Fast Food	0.01	0.00	0.00	1.38	0.89

Notes: Continued on following page.

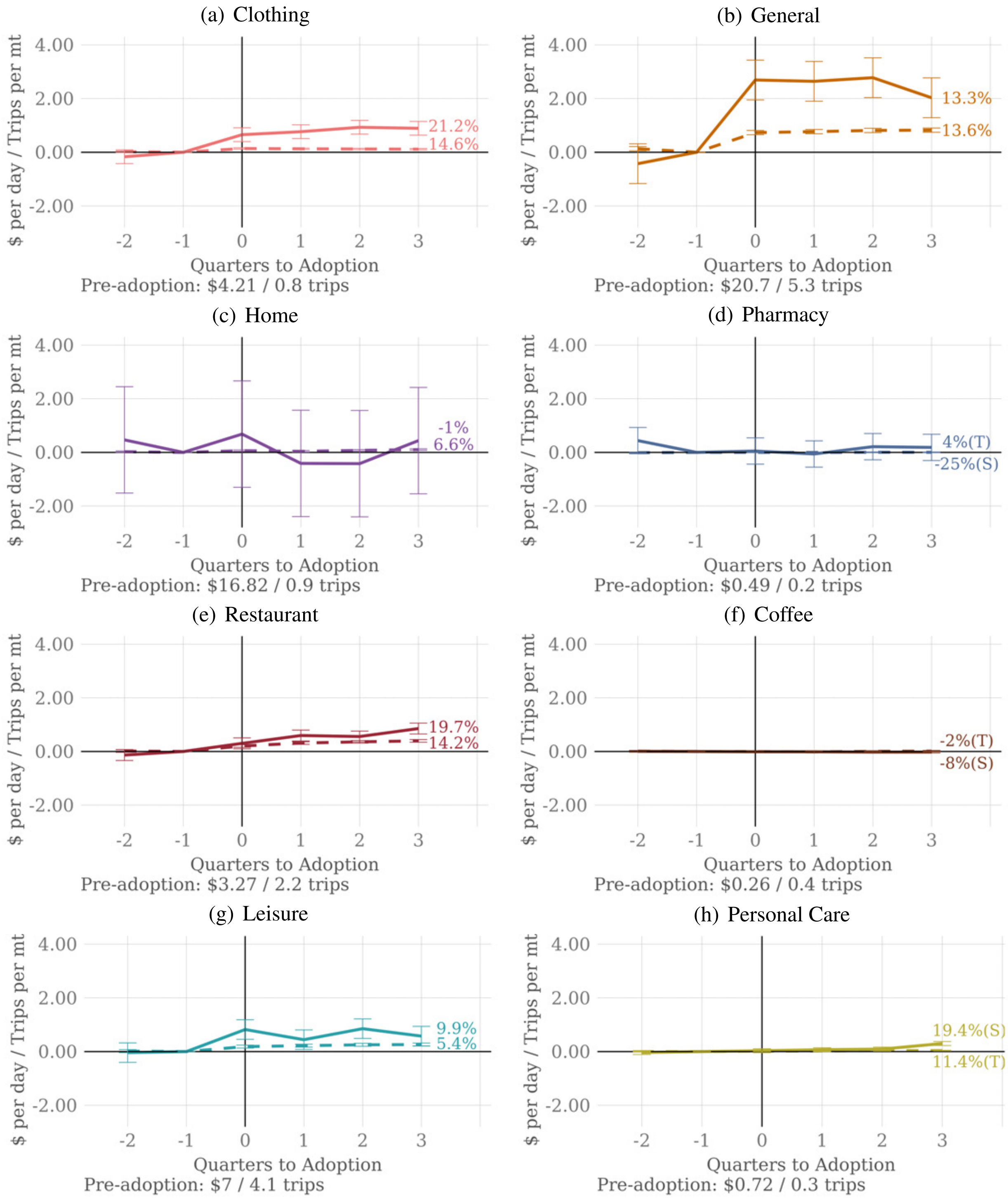
Table C3: Balance on Covariates Not Used in Matching, continued

		Means			*t-stats*	
Category	Covariate	Adopters	Non-adopters		Non-adopters	
			(All)	(Matched)	(All)	(Matched)
Restaurant Trips 6MΔ	Caterer	0.00	0.00	0.00	-1.42	0.11
	Eating Place	-0.02	-0.01	0.01	-0.85	-1.21
	Drinking Place	0.00	0.00	0.00	0.27	0.78
	Fast Food	0.00	-0.02	-0.01	1.38	1.23
Online Spending 3M	Leisure	6.72	2.51	5.27	16.21	4.28
	Pharmacy	0.43	0.32	0.95	2.16	-1.16
	Personal Care	0.73	0.20	0.51	17.02	3.45
	Coffee	0.27	0.10	0.17	5.40	1.99
Online Trips 3M	Leisure	4.09	1.64	3.05	53.82	15.85
	Pharmacy	0.20	0.12	0.16	8.59	3.44
	Personal Care	0.27	0.08	0.18	19.46	6.31
	Coffee	0.45	0.27	0.36	9.49	2.25
Online Spending 6MΔ	Leisure	0.05	0.01	0.01	0.57	-0.73
	Pharmacy	0.00	0.00	0.11	-0.62	-0.84
	Personal Care	0.03	0.00	0.02	2.93	1.18
	Coffee	0.00	0.00	0.00	-0.39	-1.22
Online Trips 6MΔ	Leisure	0.02	0.00	0.02	3.28	-0.48
	Pharmacy	0.00	0.00	0.00	1.32	0.70
	Personal Care	0.01	0.00	0.00	2.64	1.24
	Coffee	0.01	0.01	0.00	0.76	-0.10

Notes: Continued from previous page. Mean differences in spending and trips in levels and changes between early adopters and non-adopters before and after matching. Variables in this table are not included in the matching exercise. Levels are 3-month averages prior to adoption and changes are average month-to-month changes in the six months prior to adoption. Two-sided t-stats for the differences in means before and after matching are reported.
Source: Author's calculations using 3,805 early adopters of online grocery platforms, 2,226,224 non-adopters as potential matches, and 7,613 non-adopters matched as controls.

8.B Additional Results

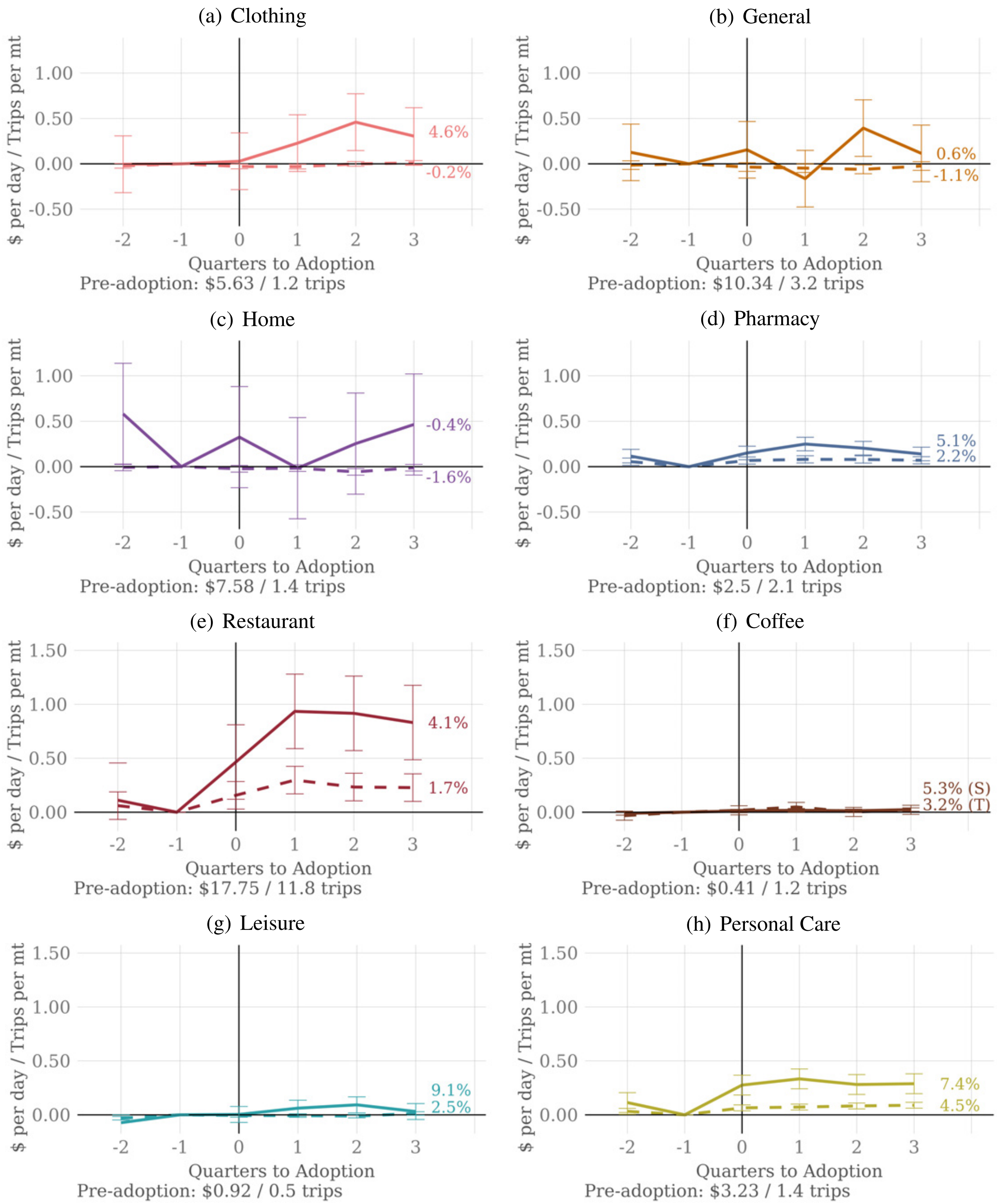
Figure C3: Online Goods and Services Spending and Trips at Adoption



Notes: Quarterly changes in online spending (solid lines) and trips (dashed lines) for non-grocery products before and after online grocery platform adoption by early adopters. Pre-adoption quarterly averages are below each panel. Series end displays the average percent change post-adoption relative to the pre-adoption quarter.

Source: Author's calculations using the transactions of 11,418 early adopters of online grocery platforms and their matched controls from a set of platform non-users.

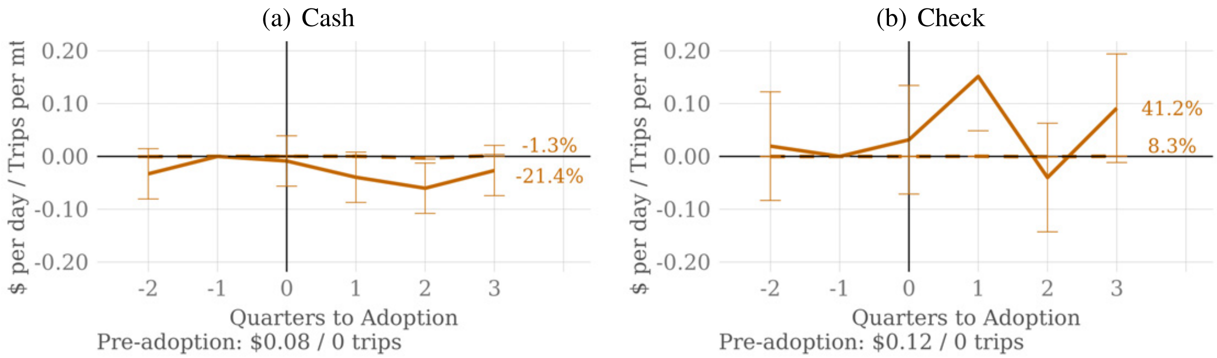
Figure C4: Offline Goods and Services Spending and Trips at Adoption



Notes: Quarterly changes in offline spending (solid lines) and trips (dashed lines) for non-grocery products before and after online grocery platform adoption by early adopters. Pre-adoption quarterly averages are below each panel. Series end displays the average percent change post-adoption relative to the pre-adoption quarter.

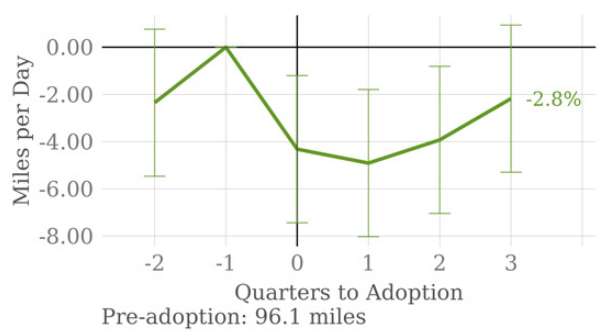
Source: Author's calculations using the transactions of 11,418 early adopters of online grocery platforms and their matched controls from a set of platform non-users.

Figure C5: Alternative Payment Instruments at Adoption



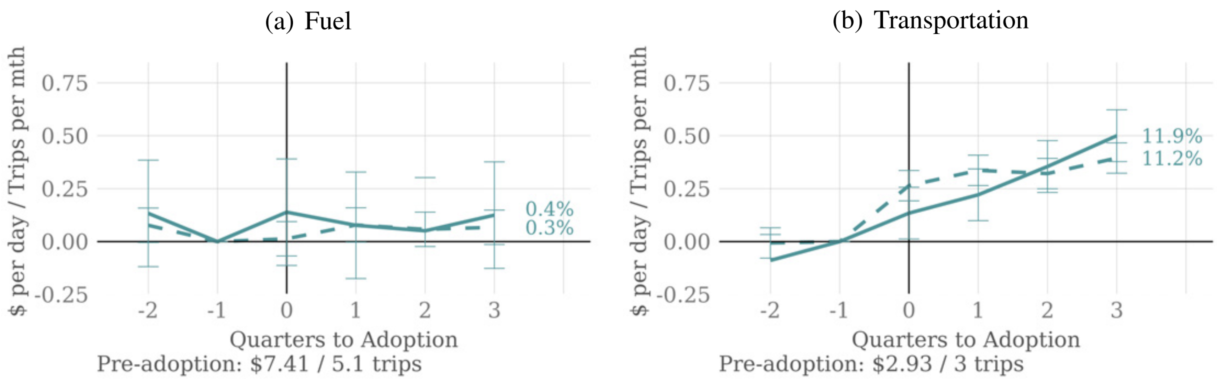
Notes: Quarterly changes in non-card payment spending (solid lines) and trips (dashed lines) before and after online grocery platform adoption by early adopters. Pre-adoption quarterly averages are below each panel. Series end displays the average percent change post-adoption relative to the pre-adoption quarter.
Source: Author's calculations using the transactions of 11,418 early adopters of online grocery platforms and their matched controls from a set of platform non-users.

Figure C6: Cumulative Grocery Miles at Adoption



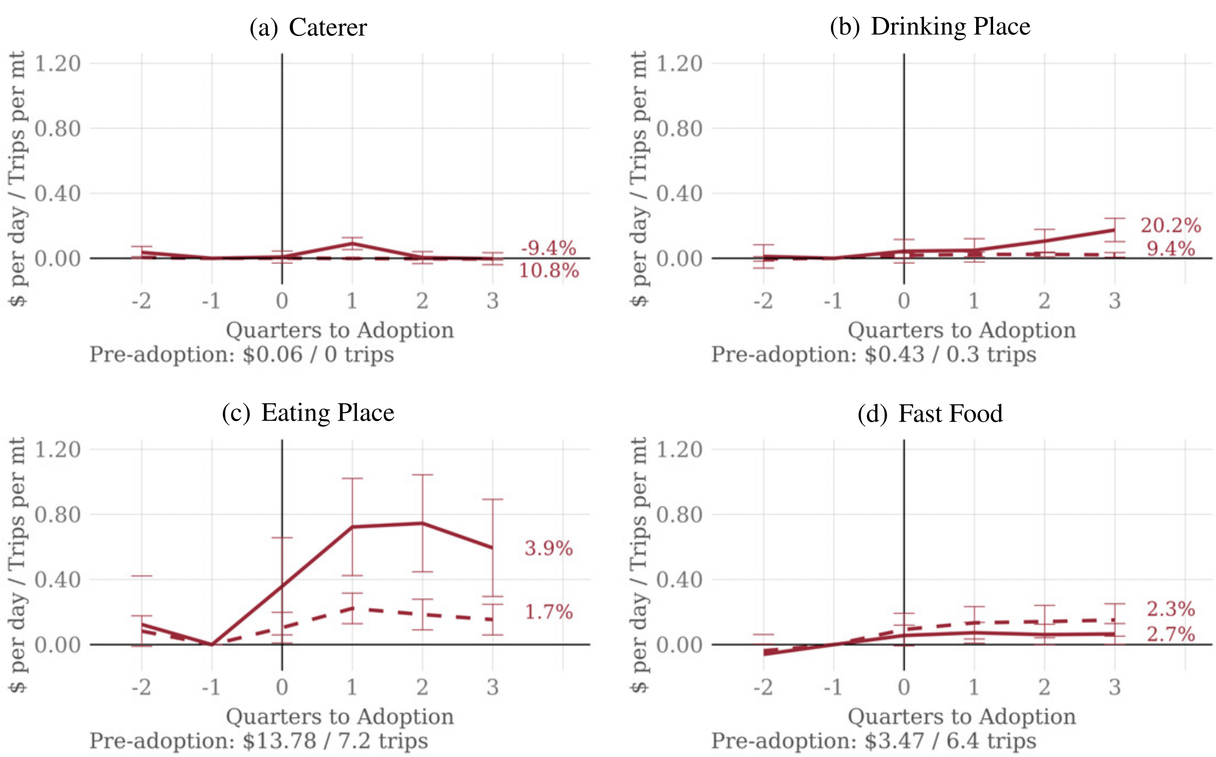
Notes: Quarterly change in single round-trip miles traveled per month for offline groceries before and after online grocery platform adoption by early adopters. Pre-adoption quarterly average below the panel. Series end displays the average percent change post-adoption relative to the pre-adoption quarter.
Source: Author's calculations using the transactions of 11,418 early adopters of online grocery platforms and their matched controls from a set of platform non-users.

Figure C7: Local Travel Spending and Trips at Adoption



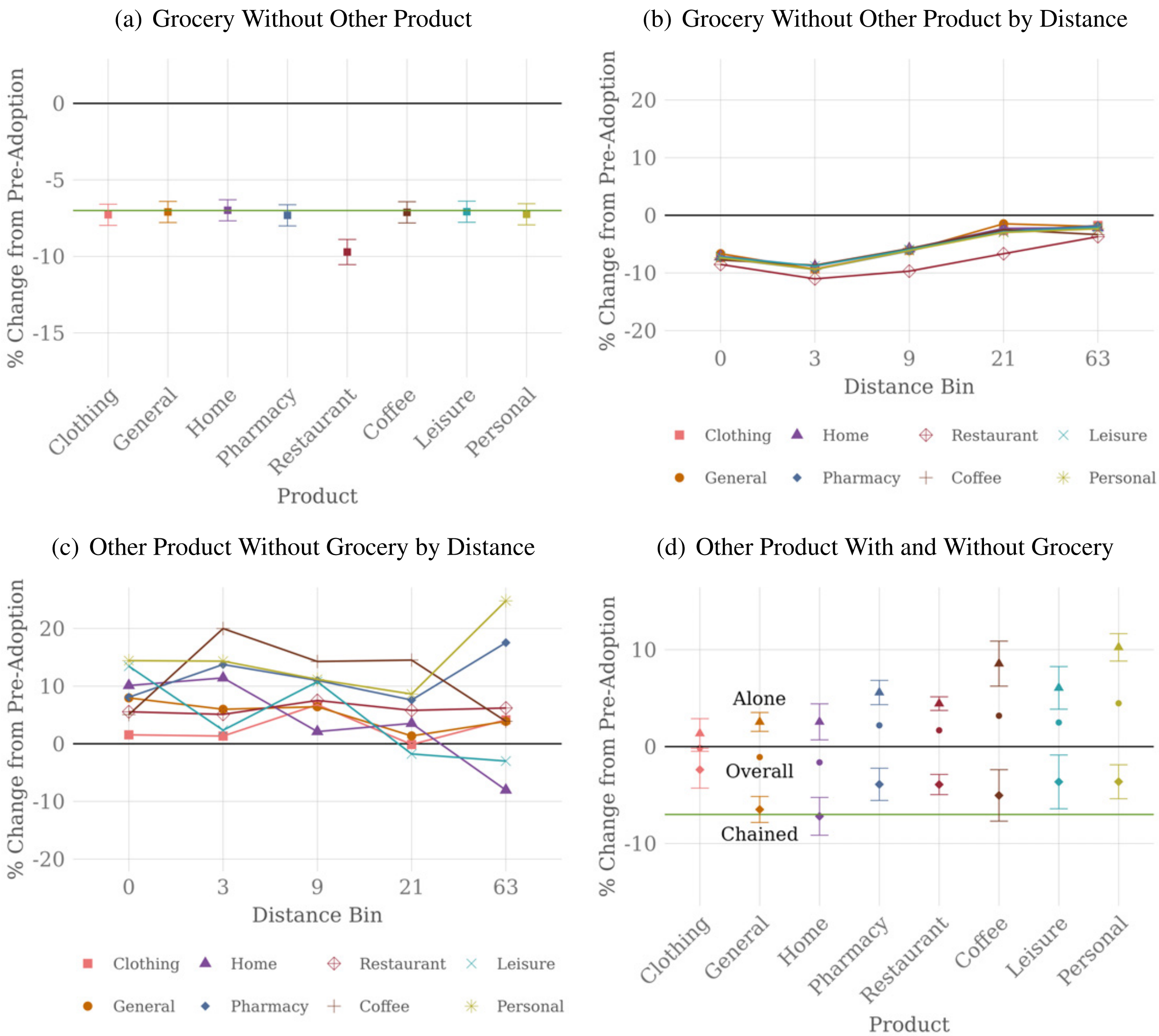
Notes: Quarterly changes in local travel spending (solid lines) and trips (dashed lines) before and after online grocery platform adoption by early adopters. Pre-adoption quarterly averages are below each panel. Series end displays the average percent change post-adoption relative to the pre-adoption quarter.
Source: Author's calculations using the transactions of 11,418 early adopters of online grocery platforms and their matched controls from a set of platform non-users.

Figure C8: Spending and Trips by Restaurant Category at Adoption



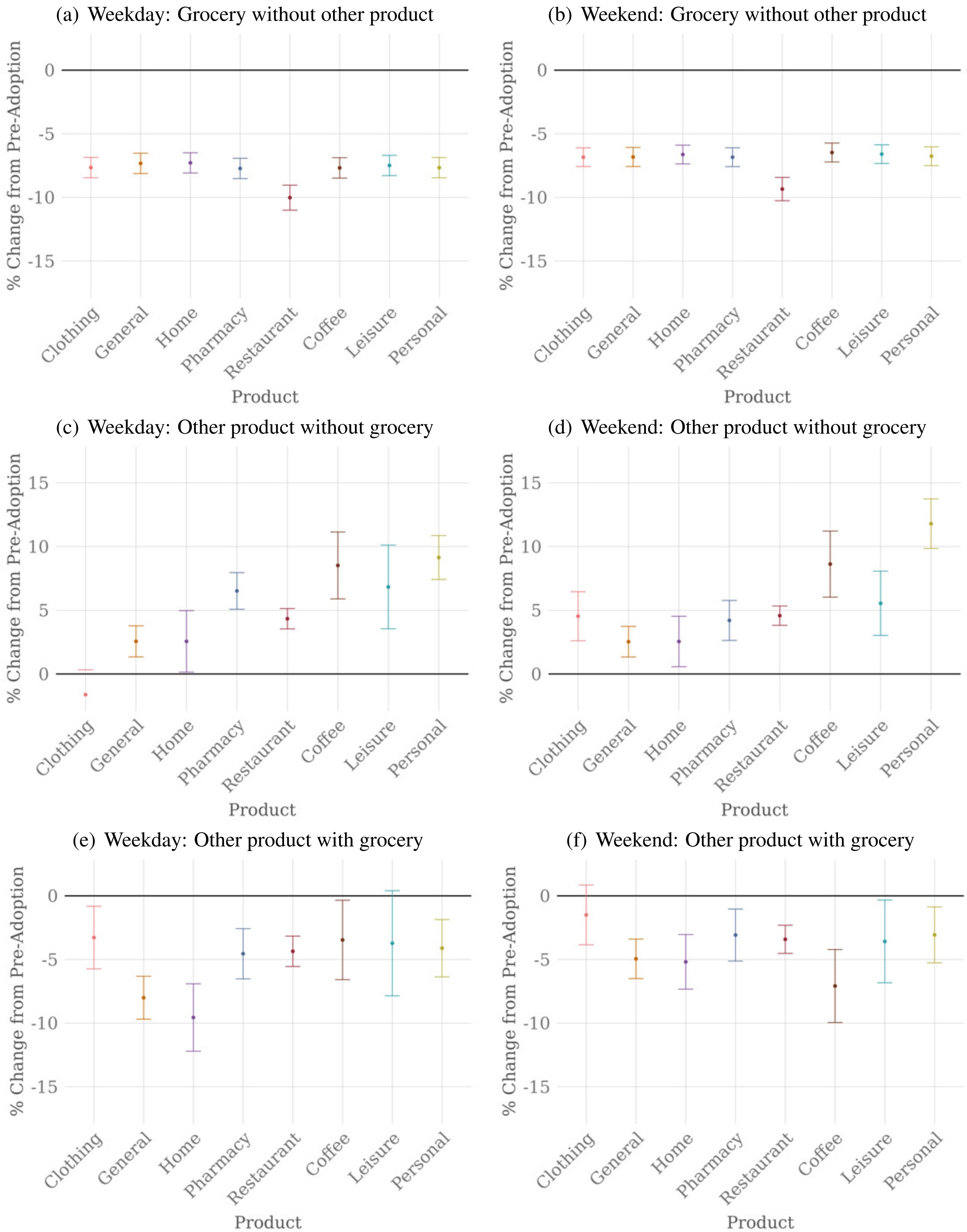
Notes: Quarterly changes in offline restaurant spending (solid lines) and trips (dashed lines) before and after online grocery platform adoption by early adopters. Pre-adoption quarterly averages are below each panel. Series end displays the average percent change post-adoption relative to the pre-adoption quarter.
Source: Author's calculations using the transactions of 11,418 early adopters of online grocery platforms and their matched controls from a set of platform non-users.

Figure C9: Changes in Store Combinations on Trips



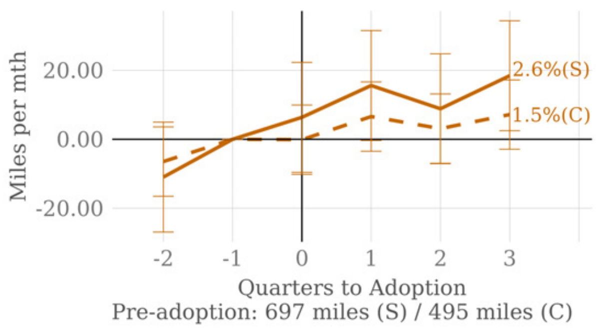
Notes: Percent change in trips for products bundled with and without grocery after online grocery platform adoption for early adopters. Trips are defined as all offline transactions that take place on the same day. Panel (a) shows the effect for grocery store alone trips without each product. Panel (b) shows the grocery store alone while panel (c) shows other products alone without grocery at each binned distance. Bin 0 corresponds to transactions that take place in the home ZIP code of the customer. For remaining bins, the upper bound of the bin displayed, e.g the second bin is for trips that are more than zero miles and less than or equal to three miles from the customer, where miles are ZIP centroid to centroid measurements. Panel (d) shows the percent change for each product trip bundled with grocery (diamonds) and without grocery (triangles) relative to the overall effect for that product (circles). Panels (a) and (d) show the overall grocery trip percent change as a horizontal green line for reference.
Source: Author's calculations using the transactions of 11,418 early adopters of online grocery platforms and their matched controls from a set of platform non-users.

Figure C10: Weekday and Weekend Effects



Notes: Percent change in trips for products bundled with and without grocery after online grocery platform adoption for early adopters split by weekday versus weekend.
Source: Author's calculations using the transactions of 11,418 early adopters of online grocery platforms and their matched controls from a set of platform non-users.

Figure C11: Miles Traveled Measured as Single v. Chained Trips



Notes: Quarterly change in single (S) versus chained (C) round-trip miles traveled per month offline before and after online grocery platform adoption for early adopters. Trips include both everyday and other products. Pre-adoption quarterly averages are below each panel. Series end displays the average percent change post-adoption relative to the pre-adoption quarter.

Source: Author's calculations using the transactions of 11,418 early adopters of online grocery platforms and their matched controls from a set of platform non-users.

Table C4: Comparison with Match Group FE Specification

Grocery Spending				
	Offline		Online	
	(1)	(2)	(3)	(4)
EA	2.91	3.14	0.41	0.51
	(0.30)	(0.24)	(0.15)	(0.14)
EA * D = -2	0.27	0.27	0.08	0.08
	(0.43)	(0.33)	(0.21)	(0.19)
EA * D = 0	-2.96	-2.96	7.16	7.16
	(0.43)	(0.33)	(0.21)	(0.19)
EA * D = 1	-2.55	-2.55	6.68	6.68
	(0.43)	(0.33)	(0.21)	(0.19)
EA * D = 2	-2.36	-2.36	6.22	6.22
	(0.43)	(0.33)	(0.21)	(0.19)
EA * D = 3	-2.20	-2.20	6.09	6.09
	(0.43)	(0.33)	(0.21)	(0.19)
Constant	19.94		1.70	
	(0.17)		(0.09)	
Match Group FE	N	Y	N	Y
R^2	0.002	0.41	0.05	0.28
Adjusted R^2	0.002	0.40	0.05	0.26

Notes: This table compares the results of estimating equation (2) for online and offline grocery spending with the inclusion of match-group fixed effects. Coefficients on Quarter *D* indicators are not reported.
Source: Author's calculations using the transactions of 11,418 early adopters of online grocery platforms and their matched controls from a set of platform non-users

9 Appendix: Model

9.A CES Framework for Consumer Welfare from Online Grocery Platforms

The model from Dolfen et al. (2023) provides a simple method for characterizing the effective price reduction and welfare gains via variety and quality implied by the increased spending across new products for early adopters of online grocery platforms using the reduced-form results. Assume consumers' utility function is CES and merchants are perfectly competitive, but vary in their offline and online production fundamentals. Adapting their same notation, in equilibrium at time t,

$$\frac{p_o^t x_o^t}{p_b^t x_b^t} = \left(\frac{q_o^t \,/\, p_o^t}{q_b^t \,/\, p_b^t}\right)^{\sigma-1}, \tag{13}$$

where price, p, quantity, x, and quality, q are indexed by online (subscript o) and offline (subscript b) and σ is the elasticity of substitution between online and offline merchants estimated to be 4.3. Take the quarter before and the year after online grocery platform adoption for early adopters as $t = 1,2$ and assume that the quality of the online and offline products and the price of the offline products do not vary across the two periods. We can derive an expression for the implied equivalent change in the price of online retail from the entry of online grocery platforms with the observed changes in total expenditures from Figure 3 of

$$\Delta p_o = \frac{p_o^2 - p_o^1}{p_o^1} = \left(\frac{p_o^1 x_o^1}{p_o^2 x_o^2} \times \frac{p_b x_b^2}{p_b x_b^1}\right)^{\left(\frac{1}{\sigma-1}\right)} - 1 = \left(\frac{128}{143} \times \frac{100}{101}\right)^{\left(\frac{1}{3.3}\right)} - 1 = -3.1\%. \tag{14}$$

The model-implied increase in consumer surplus is directly connected to the increasing share of online retail, s_o, as

$$\Delta CS = \left(\frac{1 - s_o^1}{1 - s_o^2}\right)^{\frac{\phi-1}{\phi}\frac{1}{\sigma-1}} - 1 = \left(\frac{0.56}{0.59}\right)^{\frac{0.75}{1.75}\frac{1}{3.3}} - 1 = 0.9\%, \tag{15}$$

where $\phi = 1.75$ is the parameter that captures the fixed costs of visiting merchants.

9.B Substitutions in Trip Choice

To illustrate the need for a trip choice model with rich substitution patterns, first consider the log relative probability for the trip for the restaurant only versus the trip to the general goods store only under the classic logit assumption,

$$\ln\left(\frac{P_{010}}{P_{001}}\right) = R - N + \tau \ln(d(0,1,0)/d(0,0,1)). \tag{16}$$

This depends only on the value of the restaurant and general goods store and the relative distance costs of visiting the restaurant alone versus the general goods store alone. Imagine then, the effect on this relative probability when the value of the grocery store falls once someone becomes an online grocery platform adopter. Under the logit, $-\partial \frac{P_{010}}{P_{001}} / \partial G = 0$. However, we know from the empirical evidence in section 3.D that there is a differential response for restaurants versus general goods. A model that accounts for these substitution patterns must allow for correlations in trip utilities while incorporating individual-specific distance costs.

In contrast, in the PCL model, this relative probability is

$$\ln\left(\frac{P_{010}}{P_{001}}\right) = \ln \sum_{grn' \neq 010} V_{010|010,grn'} - \ln \sum_{grn' \neq 001} V_{001|001,grn'}, \tag{17}$$

where

$$V_{grn|grn,grn'} = \exp\left(\frac{V_{grn}}{1-\sigma_{grn,grn'}} - \sigma_{grn,grn'} I_{grn,grn'}\right)$$

acts as a pair-weighted trip value. Through the trip pair inclusive values, $I_{grn,grn'}$, in each of the $V_{grn|grn,grn'}$ terms, the elasticities of the relative probability of the two trips are functions of trip utility parameters unrelated to the restaurant alone or general goods store alone trips: the grocery store value, the distance costs of all trips, and the fixed benefits of chained trips. The effects are scaled by the similarity of each pair, σ, and the utility effect of distance, τ. Whether the consumer goes relatively more to the restaurant alone or the general goods store alone when the grocery store value falls depends on the combined effects of differential changes to the pair weighted utilities for the restaurant alone and general goods store alone trips.

Table D1 shows trip utility parameters from a logit model for comparison with the PCL estimates in Table 1. Under the logit, the density gradients for the general goods store alone, the restaurant alone, and the restaurant and general goods chain are all negative, implying that more urban consumers have lower preferences for these trips. This conflicts with evidence that more urban consumers have stronger restaurant preferences (e.g. Su (2022)). This pattern is a direct result from the IIA restriction in the logit – it forces the relative preference of more and less urban consumers to change in the same proportion across any pair of these trips. In this case, the logit matches observed frequencies by making all three density gradients negative and compensating with a much higher relative value for the restaurant and a lower fixed benefit for the restaurant and general goods chain. The PCL relaxes these restrictions and produces more intuitive density patterns in which more urban consumers prefer the restaurant trip and less urban consumers prefer the general goods trip, either alone or chained with restaurants. Realistically capturing these complex substitution patterns is critical for modeling the substantial gains in welfare and trip frequency in urban areas as a result of the growth of the online grocery delivery market.

Finally, while models employing random coefficients also loosen the independence of irrelevant alternatives assumption, they fail to effectively capture the nuanced correlations present in trip choices. Compiani (2022) demonstrates that this is particularly true in situations that involve complementary choices.

9.C Utility Cost of Distance

In section 4.A, the model for daily trip choice by consumers is detailed in equation 4. I employ the logarithm of distance to introduce concavity into the straight-line distance costs. This approach is based on the realistic assumption that longer travel distances encourage consumers to opt for faster transportation modes. I also tested the model using the inverse hyperbolic sine of distance. This alternative approach, while avoiding the unrealistic implication that distances under one mile increase utility, yielded results that were both qualitatively and quantitatively consistent with the primary model.[21]

[21] In situations where distance costs are already measured in travel time, as demonstrated in studies like Miyauchi, Nakajima and Redding (2025) and Oh and Seo (2023), distance costs that scale linearly and are measured as semi-elasticities may be more appropriate.

Table D1: Logit Trip Utility Parameters

(a) Store Values

	Pre-adoption	*Post-adoption*
G	-2.22	-2.33
R	1.40	1.43
N	-3.51	-3.50
$b(110)$	0.58	0.52
$b(101)$	1.12	1.00
$b(011)$	0.65	0.68
$b(111)$	1.94	1.86

(b) Distance Coefficient

τ	0.07

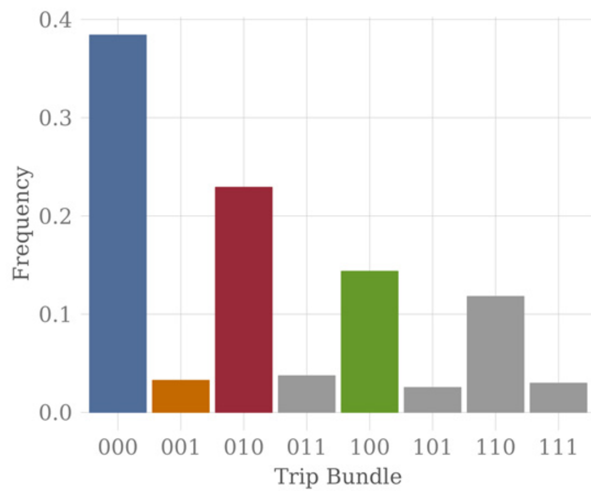
(c) Density and Socioeconomic Shifters

Trip grn =	*001*	*010*	*011*	*100*	*101*	*110*	*111*
ln Density	-0.01	-0.01	-0.07	0.03	-0.02	-0.03	-0.09
ln Age	0.20	-0.60	-0.25	0.13	0.35	-0.37	-0.07
Sex F	-0.005	-0.36	-0.31	-0.22	-0.20	-0.48	-0.46
Sex U	0.01	-0.28	-0.20	-0.12	-0.11	-0.33	-0.30
Income 2	0.10	0.26	0.38	0.19	0.33	0.45	0.59
Income 3	0.23	0.42	0.64	0.34	0.60	0.75	0.99
Income U	0.09	0.18	0.23	0.20	0.34	0.37	0.48
Asian	-0.04	0.01	-0.09	0.13	0.05	0.10	0.01
Hispanic	0.04	0.13	0.16	0.13	0.12	0.20	0.22
White	0.06	0.07	0.11	0.27	0.27	0.25	0.27
Race U	0.07	0.02	0.06	0.19	0.27	0.16	0.22

Notes: Trip utility parameters mapped from estimates for a logit model with trip utilities as specified in equation 7. Table D2 shows the estimates with standard errors. Pre-adoption values of stores derive from the trip value for that store alone, e.g. $N = \beta_0(001)$. Fixed benefits of trip chains pre-adoption derive from differences in values of standalone store visits and chained trips, e.g. $b(110) = \beta_0(110) - G - R$. Post adoption values incorporate β_3 coefficients.
Source: Author's calculations using the transactions of 6,764 early adopters of online grocery platforms and their matched controls.

9.D Additional Figures and Tables

Figure D1: Model Trip Frequencies



Notes: Trip frequencies for the eight trip types among customers used in the estimation of the PCL model of trip choice. The eight possible trip choices $\{g, r, n\}$ as $grn \in \{000, 001, 010, 011, 100, 101, 110, 111\}$ where g, r, and n are the binary choice of grocery store, restaurant, and general goods store, respectively. The none trip and each store alone are highlighted as colored bars and chained trips as grey bars.
Source: Author's calculations using the transactions of 6,764 early adopters of online grocery platforms and their matched controls from a set of platform non-users.

Table D2: Trip Choice Model Estimation Results

Trip Values			*PCL Similarities*			
	PCL	Logit				
ln Median Distance	0.04	0.07				
	(0.00)	(0.00)				
$\beta_0(001)$	-4.20	-3.51	$1\text{-}\sigma_{000,001}$	2.40	$1\text{-}\sigma_{000,100}$	1
	(0.60)	(0.05)		(0.42)		
$\beta_0(010)$	0.06	1.40	$1\text{-}\sigma_{000,010}$	0.24	$1\text{-}\sigma_{010,101}$	1.02
	(0.04)	(0.02)		(0.02)		(1.79)
$\beta_0(011)$	-2.94	-1.46	$1\text{-}\sigma_{000,011}$	0.73	$1\text{-}\sigma_{010,110}$	1.79
	(0.70)	(0.05)		(1.75)		(0.39)
$\beta_0(100)$	-2.04	-2.22	$1\text{-}\sigma_{000,101}$	2.80	$1\text{-}\sigma_{010,111}$	1.38
	(0.10)	(0.03)		(0.88)		(0.70)
$\beta_0(101)$	-6.58	-4.61	$1\text{-}\sigma_{000,110}$	0.53	$1\text{-}\sigma_{011,100}$	1.70
	(2.20)	(0.06)		(0.21)		(0.60)
$\beta_0(110)$	-1.43	-0.23	$1\text{-}\sigma_{000,111}$	0.66	$1\text{-}\sigma_{011,101}$	0.83
	(0.11)	(0.03)		(2.58)		(1.77)
$\beta_0(111)$	-3.93	-2.39	$1\text{-}\sigma_{001,010}$	0.60	$1\text{-}\sigma_{011,110}$	1.64
	(0.73)	(0.05)		(0.83)		(1.11)
$\beta_0(001)$: Post:EA	0.03	0.01	$1\text{-}\sigma_{001,011}$	0.19	$1\text{-}\sigma_{011,111}$	2.57
	(0.02)	(0.01)		(0.09)		(1.45)
$\beta_0(010)$: Post:EA	0.03	0.04	$1\text{-}\sigma_{001,100}$	2.14	$1\text{-}\sigma_{100,101}$	2.61
	(0.01)	(0.01)		(0.56)		(1.14)
$\beta_0(011)$: Post:EA	0.09	0.02	$1\text{-}\sigma_{001,101}$	0.69	$1\text{-}\sigma_{100,110}$	0.17
	(0.03)	(0.01)		(1.47)		(0.02)
$\beta_0(100)$: Post:EA	-0.11	-0.11	$1\text{-}\sigma_{001,110}$	0.26	$1\text{-}\sigma_{100,111}$	1.18
	(0.01)	(0.01)		(0.78)		(0.79)
$\beta_0(101)$: Post:EA	-0.18	-0.12	$1\text{-}\sigma_{001,111}$	0.32	$1\text{-}\sigma_{101,110}$	0.74
	(0.05)	(0.01)		(0.20)		(5.42)
$\beta_0(110)$: Post:EA	-0.06	-0.07	$1\text{-}\sigma_{010,011}$	1.63	$1\text{-}\sigma_{101,111}$	0.08
	(0.01)	(0.01)		(0.68)		(0.03)
$\beta_0(111)$: Post:EA	-0.06	-0.07	$1\text{-}\sigma_{010,100}$	0.37	$1\text{-}\sigma_{110,111}$	1.64
	(0.03)	(0.01)		(0.09)		(1.11)

Notes: Continued on following page.

Table D2: Trip Choice Model Estimation Results, continued

Density & Age			*Sex*		
	PCL	Logit		PCL	Logit
ln Density: 001	-0.05	-0.01	Female: 001	-0.03	0.00
	(0.01)	(0.00)		(0.04)	(0.01)
ln Density: 010	0.01	-0.01	Female: 010	-0.38	-0.36
	(0.00)	(0.00)		(0.01)	(0.00)
ln Density: 011	-0.06	-0.07	Female: 011	-0.37	-0.31
	(0.01)	(0.00)		(0.03)	(0.01)
ln Density: 100	0.04	0.03	Female: 100	-0.22	-0.22
	(0.00)	(0.00)		(0.01)	(0.00)
ln Density: 101	-0.06	-0.02	Female: 101	-0.31	-0.20
	(0.02)	(0.00)		(0.06)	(0.01)
ln Density: 110	0.01	-0.03	Female: 110	-0.46	-0.48
	(0.00)	(0.00)		(0.01)	(0.00)
ln Density: 111	-0.07	-0.09	Female: 111	-0.48	-0.46
	(0.02)	(0.00)		(0.05)	(0.01)
ln Age: 001	0.36	0.20	Sex U: 001	0.04	0.01
	(0.06)	(0.01)		(0.02)	(0.01)
ln Age: 010	-0.24	-0.60	Sex U: 010	-0.21	-0.28
	(0.01)	(0.01)		(0.01)	(0.00)
ln Age: 011	0.09	-0.25	Sex U: 011	-0.12	-0.20
	(0.05)	(0.01)		(0.02)	(0.01)
ln Age: 100	0.08	0.13	Sex U: 100	-0.01	-0.12
	(0.01)	(0.01)		(0.01)	(0.01)
ln Age: 101	0.53	0.35	Sex U: 101	0.06	-0.11
	(0.15)	(0.01)		(0.04)	(0.01)
ln Age: 110	-0.16	-0.37	Sex U: 110	-0.17	-0.33
	(0.02)	(0.01)		(0.01)	(0.01)
ln Age: 111	0.13	-0.07	Sex U: 111	-0.16	-0.30
	(0.06)	(0.01)		(0.03)	(0.01)

Notes: Continued on following page.

Table D2: Trip Choice Model Estimation Results, continued

Income					
	PCL	Logit		PCL	Logit
Income 2: 001	0.10	0.10	Income U: 001	0.15	0.09
	(0.04)	(0.01)		(0.04)	(0.01)
Income 2: 010	0.20	0.26	Income U: 010	0.12	0.18
	(0.01)	(0.01)		(0.01)	(0.01)
Income 2: 011	0.29	0.38	Income U: 011	0.19	0.23
	(0.05)	(0.01)		(0.05)	(0.01)
Income 2: 100	0.21	0.19	Income U: 100	0.26	0.20
	(0.01)	(0.01)		(0.01)	(0.01)
Income 2: 101	0.54	0.33	Income U: 101	0.65	0.34
	(0.12)	(0.01)		(0.15)	(0.01)
Income 2: 110	0.51	0.45	Income U: 110	0.47	0.37
	(0.03)	(0.01)		(0.03)	(0.01)
Income 2: 111	0.75	0.59	Income U: 111	0.66	0.48
	(0.11)	(0.01)		(0.10)	(0.01)
Income 3: 001	0.29	0.23			
	(0.05)	(0.01)			
Income 3: 010	0.30	0.42			
	(0.01)	(0.01)			
Income 3: 011	0.49	0.64			
	(0.06)	(0.01)			
Income 3: 100	0.33	0.34			
	(0.02)	(0.01)			
Income 3: 101	0.98	0.60			
	(0.22)	(0.01)			
Income 3: 110	0.69	0.75			
	(0.03)	(0.01)			
Income 3: 111	0.99	0.99			
	(0.13)	(0.01)			

Notes: Continued on following page.

Table D2: Trip Choice Model Estimation Results, continued

Race					
	PCL	Logit		PCL	Logit
Asian: 001	-0.11	-0.04	White: 001	-0.01	0.06
	(0.06)	(0.02)		(0.05)	(0.02)
Asian: 010	0.19	0.01	White: 010	0.20	0.07
	(0.01)	(0.01)		(0.01)	(0.01)
Asian: 011	0.06	-0.09	White: 011	0.17	0.11
	(0.05)	(0.02)		(0.05)	(0.02)
Asian: 100	0.27	0.13	White: 100	0.35	0.27
	(0.02)	(0.01)		(0.02)	(0.01)
Asian: 101	0.46	0.05	White: 101	0.61	0.27
	(0.12)	(0.03)		(0.14)	(0.02)
Asian: 110	0.44	0.10	White: 110	0.50	0.25
	(0.02)	(0.01)		(0.03)	(0.01)
Asian: 111	0.41	0.01	White: 111	0.58	0.27
	(0.07)	(0.03)		(0.09)	(0.02)
Hispanic: 001	-0.06	0.04	Race U: 001	0.06	0.07
	(0.05)	(0.02)		(0.04)	(0.02)
Hispanic: 010	0.34	0.13	Race U: 010	0.14	0.02
	(0.01)	(0.01)		(0.01)	(0.01)
Hispanic: 011	0.36	0.16	Race U: 011	0.17	0.06
	(0.05)	(0.02)		(0.04)	(0.02)
Hispanic: 100	0.11	0.13	Race U: 100	0.24	0.19
	(0.02)	(0.01)		(0.02)	(0.01)
Hispanic: 101	0.15	0.12	Race U: 101	0.67	0.27
	(0.06)	(0.03)		(0.17)	(0.02)
Hispanic: 110	0.50	0.20	Race U: 110	0.39	0.16
	(0.02)	(0.01)		(0.02)	(0.01)
Hispanic: 111	0.70	0.22	Race U: 111	0.53	0.22
	(0.10)	(0.02)		(0.08)	(0.02)

Notes: Coefficient estimates of equation 7 via maximum likelihood. Standard errors in parentheses. Dependent variable is the trip choice from the three-store bundle combination of grocery store, restaurant, and general goods store, in which trip choice $\{g,r,n\}$ corresponds to the binary combination of each store, respectively, and can take values $grn \in \{000,001,010,011,100,101,110,111\}$. The trip value for trip 000 and $\sigma_{000,100}$ are normalized to 0. Coefficients on *Post* and *EA* are not reported. For the PCL and logit models, the log-likelihoods are $-7,552,687$ and $-7,542,846$, McFadden's pseudo R^2 values are 0.006 and 0.007, and likelihood ratio test statistics are 85,121 (df = 134) and 104,804 (df = 106), respectively.
Source: Author's calculations using 4,553,427 customer-day observations for 6,764 early adopters of online grocery platforms and their matched controls drawn from platform non-users.

Table D3: Effects of Mature Online Grocery Platform Market on Stores

Dependent Variable: Ln Store Visits			
	Grocery	Restaurant	General
Ln Distance from Center	0.37	-0.06	-0.04
	(0.20)	(0.03)	(0.02)
Ln Density	-0.15	0.03	0.01
	(0.08)	(0.02)	(0.01)
Ln Age	-0.03	0.03	-0.01
	(0.14)	(0.01)	(0.03)
Sex F	-0.04	-0.02	-0.03
	(0.03)	(0.004)	(0.01)
Sex U	0.05	-0.01	-0.02
	(0.06)	(0.01)	(0.01)
Income 2	-0.04	-0.03	-0.02
	(0.11)	(0.01)	(0.01)
Income 3	-0.13	-0.02	-0.02
	(0.25)	(0.03)	(0.03)
Income U	-0.03	-0.04	-0.03
	(0.13)	(0.01)	(0.01)
Asian	-0.23	0.04	-0.01
	(0.14)	(0.02)	(0.01)
Hispanic	-0.12	0.03	0.02
	(0.07)	(0.01)	(0.01)
White	-0.29	0.06	-0.004
	(0.23)	(0.04)	(0.02)
Race U	-0.32	0.05	0.04
	(0.21)	(0.03)	(0.03)
Mean	-0.71	0.12	0.12
Adjusted R^2	0.38	0.31	0.14

Notes: Estimates for $LnStoreVisits_{iz} = \mathbf{x}'_{zi}\boldsymbol{\beta} + \phi_c$. The vector $\mathbf{x}_{zi}$ captures the distance from city center and population density of the customer's residential ZIP code, along with their socioeconomic characteristics of age, sex, income, and race where z indexes ZIP codes and i indexes customers. The ϕ_c are CBSA fixed effects. Clustered (CBSA) standard errors in parentheses. Table 3 shows changes in specific trip types that aggregate to grocery, restaurant, and general goods visits.

Source: Author's calculations using data for 1,394,570 customers. Customers are from the balanced panel, have a complete set of representative "preferred stores", and reside in a ZIP code for which platform value is estimated and measures of distance from city center and population density are available.

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